

TICVAI

Core AI Platform & Enterprise Intelligence

Dashboard requirements and functional specification

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Core AI Platform & Enterprise Intelligence — Board 1: AI Assistant & Enterprise Knowledge

Module: Core AI Platform & Enterprise Intelligence

Board: 1 of 4

Screens: 10

Board Type: Enterprise AI Assistant / RAG / Knowledge Management / Semantic Search / Grounded Answers / Permissions



Board Purpose

This board establishes the **knowledge and conversational intelligence layer** behind the general TICVAI AI Assistant.

The objective is to allow authorized users to communicate with TICVAI naturally and ask questions about both **system data** and **approved enterprise knowledge**.

Examples:

“What was yesterday's revenue at Dubai Theme Park?”

“Compare this weekend's attendance with last weekend.”

“Which attractions are forecast to have the longest queues tomorrow?”

“Why was transaction TXN-984521 classified as high risk?”

“What is our refund policy for this ticket?”

“Show me ticket products with low sales this month.”

“How do I configure a family ticket?”

The Assistant should not simply send every question to an LLM and trust the answer.

The architecture should be:

User Question

↓

Identity + Tenant + Role + Permissions + Conversation Context

↓

Intent & Context Understanding

↓

Knowledge / Data Retrieval Layer

**TICVAI Structured Data + APIs + Approved Documents + Policies + Configuration
Metadata + Knowledge Base**

↓

RAG / Retrieval & Grounding

↓

LLM / AI Reasoning

↓

Permission & Governance Validation

↓

Grounded Response

Answer + Sources + Evidence + Confidence/Limitations + Suggested Next Step

Board 1 is primarily about:

“How does TICVAI AI know what it knows, retrieve the correct information and give users trustworthy answers?”

Actual action/tool execution comes in **Board 2**.

Purpose

Provide the main conversational interface through which users interact with TICVAI AI.

This becomes the general **TICVAI Copilot** experience.

Main Interface

Left side:

Conversation History

1. Today
2. Yesterday
3. Last 7 Days
4. Favorites
5. Shared Conversations where permitted

Main workspace:

Ask TICVAI

Example placeholder:

Ask about sales, attendance, customers, operations, forecasts, fraud, configuration or TICVAI knowledge...

Suggested Questions

Dynamic suggestions based on role.

For a CEO:

1. How is the business performing today?
2. Compare revenue with last week.
3. What requires my attention?
4. What is tomorrow's attendance forecast?

For Operations:

1. Which attractions will be busiest tomorrow?
2. Show staffing gaps.
3. Which POS locations have the longest predicted wait?
4. Show today's access issues.

For Finance:

1. Show yesterday's payment reconciliation issues.
2. What are the largest refunds this week?
3. Compare revenue by channel.

Response Structure

A response should support:

Direct Answer

- 1.

Key Findings

- 1.

Supporting Data

- 1.

Sources / Evidence

- 1.

Suggested Follow-Up

Example:

Saturday attendance is forecast at **16,120 visitors**, approximately **8.4% above the historical Saturday average**.

Then:

Sources

1. Forecast FRC-20260919-v7
2. Current Bookings
3. Historical Attendance

Rich Response Components

Support:

1. Text
2. KPI
3. Table

4. Chart
5. Trend
6. Comparison
7. Alert
8. Link to TICVAI screen
9. Source citation

Conversation Controls

1. New Conversation
2. Rename
3. Pin
4. Archive
5. Clear Context
6. Share where permitted
7. Export where permitted

Actions

Ask | Regenerate | View Sources | Open Related Screen | Continue Analysis

Screen 2 — Assistant Profile, Role & Behavior Configuration

Purpose

Configure how the TICVAI Assistant behaves for different tenants, roles and business contexts.

The Assistant should not behave identically for every user.

Assistant Profile

Fields:

1. Assistant Name
2. Tenant
3. Business Unit
4. Venue

5. Supported Languages
6. Default Language
7. Tone
8. Response Detail
9. Default Context
10. Knowledge Profile
11. Tool Profile
12. Governance Profile
13. Status

Example Profiles

Executive Assistant

Focus:

1. Revenue
2. Attendance
3. Forecast
4. Business Performance
5. Risks
6. Strategic Exceptions

Operations Assistant

Focus:

1. Capacity
2. Queue
3. Access
4. Workforce
5. POS
6. Resources

Finance Assistant

Focus:

1. Revenue
2. Payments
3. Refunds
4. Reconciliation
5. Settlement

Role Awareness

The Assistant should know:

Who is asking?

not just:

What was asked?

Example:

CEO asks:

“Show total revenue.”

May receive organization-level data if authorized.

Venue Manager asks the same question:

“Show total revenue.”

May receive only:

Their Venue

Response Preferences

Configure:

1. Concise
2. Standard
3. Detailed
4. Executive Summary
5. Data-First

Language

Support multilingual interaction.

Example:

User asks in Arabic.

↓

TICVAI understands intent.

↓

Retrieves English business data/policies.

↓

Responds in Arabic.

Important

Language should not change authorization.

Arabic, English or any other supported language must return the same authorized information.

Actions

Create Profile | Assign Roles | Preview | Test | Activate

Screen 3 — Enterprise Knowledge Source Registry

Purpose

Define all approved knowledge sources available to the TICVAI Assistant.

This is one of the most important backend configuration screens.

Knowledge Source Types

Support:

TICVAI Product Knowledge

1. User Guides
2. Module Documentation
3. Configuration Guides
4. Release Notes
5. API Documentation
6. Help Articles

Tenant Knowledge

1. SOPs
2. Venue Policies
3. Refund Policies
4. Operational Manuals
5. Training Documents
6. FAQs
7. Terms & Conditions

Structured TICVAI Sources

1. Ticketing
2. Sales
3. Attendance
4. Finance
5. CRM
6. Access Control
7. F&B
8. Retail
9. Inventory
10. Forecasting
11. Fraud
12. Resource Management
13. Other approved modules

External Enterprise Sources

Where supported and authorized:

1. SharePoint
2. Google Drive
3. OneDrive
4. Confluence
5. Document Repository

6. Approved Website / CMS
7. API Source

Source Record

Each source should contain:

1. Source ID
2. Name
3. Type
4. Owner
5. Tenant
6. Venue / Scope
7. Connection Type
8. Content Type
9. Security Classification
10. Permission Model
11. Sync Method
12. Sync Frequency
13. Last Sync
14. Freshness
15. Index Status
16. Status

Example

Venue Refund Policy

Type:

PDF / Policy

Owner:

Operations

Tenant:

Dubai Theme Park

Last Updated:

10 Sep 2026

Last Indexed:

10 Sep 2026 14:10

Status:

Current

Source Health

Show:

1. Connected
2. Syncing
3. Current
4. Stale
5. Failed
6. Permission Error
7. Indexing Error

Actions

Add Source | Configure | Sync Now | Reindex | Disable | View Content

Screen 4 — Knowledge Ingestion, Processing & Indexing

Purpose

Configure how enterprise content becomes searchable and usable by AI.

Adding a PDF to a folder should not automatically mean:

“The AI understands it.”

TICVAI requires a controlled ingestion pipeline.

Pipeline

Source

↓

Fetch / Sync

↓

Parse

↓

Normalize

↓

Classify

↓

Chunk

↓

Metadata Enrichment

↓

Permission Mapping

↓

Embedding

↓

Index

↓

Quality Validation

↓

Available to Assistant

Ingestion Configuration

Fields:

1. Parser
2. Content Type
3. Chunk Strategy
4. Chunk Size
5. Overlap
6. Metadata Extraction

7. Language Detection
8. Embedding Profile
9. Index
10. Permission Mapping
11. Update Strategy

Metadata

Capture where possible:

1. Document Title
2. Section
3. Page
4. Author
5. Department
6. Effective Date
7. Expiry Date
8. Version
9. Venue
10. Module
11. Product
12. Policy Type
13. Classification

Version Handling

Example:

Refund Policy v2

must supersede:

Refund Policy v1

without deleting the historical reference.

Incremental Update

If only 5% of a large source changes, TICVAI should support updating affected content rather than unnecessarily rebuilding everything.

Processing Status

1. Queued
2. Processing
3. Indexed
4. Failed
5. Needs Review
6. Superseded

Actions

Process | Reprocess | Preview Chunks | Validate | Publish to Knowledge

Screen 5 — Knowledge Permission & Access Control

Purpose

Ensure the Assistant can retrieve only information that the requesting user is authorized to access.

This is a **non-negotiable architecture requirement**.

The correct flow is not:

Retrieve everything → Ask LLM not to show restricted information.

It should be:

User Identity

↓

Tenant

↓

Role

↓

RBAC/PBAC

↓

Knowledge Permission

↓

Retrieve Only Authorized Content

Permission Dimensions

Support:

1. Tenant
2. Business Unit
3. Venue
4. Department
5. Role
6. User
7. Module
8. Data Type
9. Knowledge Source
10. Document
11. Classification

Example

Document:

Executive Financial Forecast FY2027

Allowed:

CEO

CFO

Finance Director

Not allowed:

Cashier

Gate Operator

External Reseller

Retrieval Security

Permission filtering should occur during retrieval/server-side, not only after generation.

Structured Data

The same applies to live TICVAI data.

Example:

Venue Manager asks:

“Show revenue across all venues.”

If authorized only for Venue A:

TICVAI should retrieve:

Venue A

not retrieve all venues and rely on the LLM to hide the others.

Permission Test

Administrator should be able to simulate:

User: Venue Manager

Question:

Show company revenue.

Result:

Access Limited — Venue A only

Actions

Configure Access | Test User | View Permission Map | Audit Access

Screen 6 — Enterprise Search, Retrieval & RAG Configuration

Purpose

Configure how TICVAI finds the most relevant authorized information before sending context to the model.

This is the heart of the **RAG architecture**.

What is RAG?

For Soft Labs, the concept is:

User asks question

↓

TICVAI searches approved knowledge/data

↓

Finds relevant information

↓

Provides that information to the LLM

↓

LLM generates an answer grounded in retrieved evidence.

Retrieval Modes

Support:

1. Keyword Search
2. Semantic Search
3. Hybrid Search
4. Structured Query
5. Metadata Filter
6. Vector Search
7. Reranking

Recommended:

Hybrid Retrieval

where appropriate.

Search Configuration

Fields:

1. Search Profile
2. Knowledge Sources
3. Top-K Retrieval
4. Similarity Threshold
5. Metadata Filters
6. Reranker
7. Freshness Weight
8. Authority Weight

9. Permission Filter
10. Duplicate Handling

Example Query

What is the refund policy for an unused annual pass?

TICVAI may retrieve:

1. Annual Pass Refund Policy
2. Membership Terms
3. Ticket Service Policy

Then rerank the sources.

Retrieval Preview

Show:

Result	Relevance	Authority	Freshness	Permission
Annual Pass Policy	96%	Official	Current	Allowed
Membership Terms	89%	Official	Current	Allowed
Old FAQ	72%	Support	Stale	Allowed

Stale Knowledge

The system should identify:

STALE SOURCE

rather than treat an outdated document as current.

No Good Result

If evidence is insufficient:

“I couldn't find enough approved information to answer this reliably.”

This is preferable to hallucinating.

Actions

[Test Search](#) | [View Retrieval](#) | [Tune Profile](#) | [Compare Search Methods](#)

Screen 7 — Grounding, Citations & Answer Reliability

Purpose

Control how the Assistant demonstrates the basis for its answers and prevents unsupported claims.

Response Reliability Policy

Configure when the Assistant:

1. Answers Normally
2. Answers with Warning
3. Requests Clarification
4. Shows Conflicting Sources
5. Says Information Is Unavailable
6. Refuses Due to Permission
7. Escalates to Human / Support

Citation Types

Support references to:

1. TICVAI Screen
2. Report
3. Transaction
4. Forecast
5. Fraud Assessment
6. Policy
7. Document
8. Knowledge Article
9. API Documentation
10. Configuration Object

Example

User:

Why is tomorrow expected to be busy?

Assistant:

Tomorrow's attendance forecast is 16,120 visitors. The primary contributing signals are stronger booking pace, weekend demand and the scheduled event.

Sources:

1. Forecast FRC-20260919-v7
2. Booking Pace Report
3. Event Calendar

Source Preview

Clicking a source should show:

1. Source Name
2. Relevant Section
3. Timestamp
4. Version
5. Owner
6. Freshness

subject to permissions.

Conflicting Knowledge

Example:

Policy A says:

Refund allowed within 48 hours.

Policy B says:

Refund allowed within 24 hours.

TICVAI should flag:

CONFLICTING SOURCES

instead of selecting one silently.

Confidence / Reliability

Do not create a fake universal “AI confidence” percentage.

Reliability can consider structured factors such as:

1. Retrieval Quality
2. Source Authority
3. Source Freshness

4. Number of Supporting Sources
5. Conflicts
6. Data Completeness

Actions

[View Sources](#) | [View Evidence](#) | [Report Incorrect Answer](#) | [Open Source](#)

Screen 8 — Conversational Context & Session Intelligence

Purpose

Allow natural multi-turn conversations while keeping context controlled, relevant and secure.

Example:

User:

Show Saturday's attendance forecast.

Assistant:

16,120.

User:

Compare it with Friday.

The Assistant should understand:

“it” = Saturday attendance forecast

without requiring the user to repeat everything.

Context Types

Support:

1. Current Conversation
2. Selected Venue
3. Selected Date
4. Selected Customer
5. Selected Event
6. Selected Transaction

7. Selected Ticket
8. Selected Report
9. User Role
10. Previous Question
11. Approved Session Context

Context Panel

Display:

Current Context

Venue:

Dubai Theme Park

Date:

19 Sep 2026

Topic:

Attendance Forecast

Comparison:

Previous Day

Context Controls

Allow user to:

1. View Context
2. Remove Item
3. Clear Context
4. Start New Topic

Context Boundaries

Conversation context must never override permissions.

If the user loses access to Venue B during the conversation:

Venue B information should no longer remain available simply because it was mentioned earlier.

Long Conversation Management

Support:

1. Context Summarization
2. Relevant History Selection
3. Context Window Management
4. Token Optimization

without losing important business meaning.

Cross-Conversation Context

If enabled, this must be explicitly governed.

Do not automatically expose previous sensitive conversations to unrelated sessions.

Actions

View Context | Clear Context | Pin Context | Start New Topic

Screen 9 — Knowledge Quality, Freshness & Feedback Center

Purpose

Ensure the AI Assistant's knowledge remains accurate, current and useful.

The quality of the LLM does not matter if the knowledge feeding it is outdated.

Knowledge Health KPIs

1. Active Sources
2. Indexed Documents
3. Current Sources
4. Stale Sources
5. Failed Sources
6. Permission Errors
7. Conflicting Sources
8. Unanswered Questions
9. Low-Reliability Answers
10. User-Reported Issues

Freshness Dashboard

Example:

Source	Last Updated	Last Indexed	Status
Refund Policy	10 Sep	10 Sep	Current
Operations SOP	02 Sep	02 Sep	Current
Membership FAQ	15 Jun	15 Jun	Stale
API Guide	12 Sep	12 Sep	Current

Knowledge Gap Detection

Analyze questions the Assistant could not answer.

Example:

Users asked 38 times:

“What is the procedure for damaged RFID wristbands?”

No authoritative knowledge source found.

Recommendation

Create Knowledge Article

This can become a task for the appropriate content owner.

User Feedback

Support:

1. Helpful
2. Not Helpful
3. Incorrect
4. Outdated
5. Missing Information
6. Wrong Source
7. Permission Issue

Feedback Routing

Example:

Outdated Policy

→ Knowledge Owner

Wrong System Data

→ Owning TICVAI Module

Poor AI Answer

→ AI Platform Evaluation

Important

User feedback should not directly rewrite the AI knowledge base.

It creates:

Review / Improvement Candidate

Actions

Review Stale Sources | Review Gaps | Review Feedback | Assign Owner | Reindex

Screen 10 — Assistant Testing, Evaluation & Publication

Purpose

Provide a controlled environment where Soft Labs/TICVAI administrators can test an Assistant configuration before publishing it to users.

This is critical for enterprise AI.

Test Workspace

Select:

1. Assistant Profile
2. User Role
3. Tenant
4. Venue
5. Language
6. Knowledge Profile
7. Retrieval Profile
8. Model Profile
9. Environment

Test Questions

Create evaluation sets such as:

Finance

What was yesterday's net sales?

Operations

Which attraction is expected to have the highest queue tomorrow?

Policy

Can this ticket be refunded?

Security

Show revenue for another tenant.

Expected:

ACCESS DENIED

Evaluation Dimensions

Measure:

1. Answer Correctness
2. Groundedness
3. Source Quality
4. Citation Accuracy
5. Permission Compliance
6. Retrieval Relevance
7. Response Latency
8. Completeness
9. Language Quality

Golden Test Set

Maintain approved question/expected-result scenarios.

Example:

Question:

"Can a cashier see company-wide financial results?"

Expected:

NO

Required Behavior:

Restrict answer to authorized scope.

Regression Testing

Before publishing a new:

1. Model
2. Prompt
3. Retrieval Profile
4. Knowledge Configuration

run previous tests again.

This prevents an improvement in one area from breaking another.

Compare Versions

Assistant v2.4

vs.

Assistant v2.5

Compare:

1. Grounded Answers
2. Incorrect Answers
3. Citation Quality
4. Permission Failures
5. Latency
6. Cost

Publication Workflow

Statuses:

Draft

→ **Testing**

→ **Review**

→ **Approved**

→ **Published**

→ **Superseded**

Rollback

Allow:

Rollback to Previous Assistant Version

if quality deteriorates.

Actions

Run Tests | Compare Versions | Submit Approval | Publish | Rollback

Critical Architecture — Do Not Build “Just a ChatGPT Window”

This is probably the most important message for Soft Labs.

The architecture should **not** be:

User

↓

ChatGPT API

↓

Answer

That is not an enterprise AI platform.

The recommended architecture is:

USER

|

↓

TICVAI AI ASSISTANT

|

└─ User Identity

└─ Tenant

└─ Role

└─ Permissions

└─ Language

└─ Conversation Context

|

↓

INTENT & CONTEXT LAYER

|

↓

AUTHORIZED RETRIEVAL

|

└──────────────────┴──────────────────┘

↓

↓

↓

TICVAI DATA KNOWLEDGE BASE EXTERNAL APPROVED

& APIs / RAG KNOWLEDGE

|

|

|

└──────────────────┴──────────────────┘

↓

RETRIEVAL / RERANKING

↓

GROUNDING CONTEXT

↓

LLM / AI

↓

RESPONSE VALIDATION

↓

GOVERNANCE / SECURITY

↓

GROUNDING ANSWER

|

└─ Answer

└─ Evidence

└─ Sources

└─ Limitations

└─ Suggested Next Step

Two Types of Knowledge — Critical Distinction

Soft Labs must understand that TICVAI has two fundamentally different information sources.

1. Structured Live Data

Examples:

1. Sales
2. Tickets
3. Customers
4. Attendance
5. Capacity
6. Inventory
7. Forecast
8. Fraud
9. Finance

For this type of question:

“How much revenue did we make yesterday?”

The Assistant should normally retrieve authoritative structured data through approved TICVAI APIs/services.

It should **not** search a vector database hoping to find yesterday's revenue in a document.

2. Unstructured Knowledge

Examples:

1. Policies
2. Manuals
3. SOPs
4. Documentation
5. Terms
6. FAQs

For:

“What is our refund policy?”

RAG/document retrieval is appropriate.

Therefore:

Structured Question

→ TICVAI Data/API

Knowledge Question

→ RAG

Mixed Question

→ Both.

Example of a Mixed Question

User asks:

“Why did refunds increase this week, and is this within our refund policy?”

This requires:

Live Refund Data

- 1.

Historical Comparison

- 1.

Refund Policy Knowledge

↓

AI reasoning

↓

Grounded response.

That is exactly why we need this architecture rather than a simple chatbot.

Knowledge Object

I recommend a standardized knowledge object:

KnowledgeObject

|

└─ Knowledge ID

└─ Tenant

└─ Source

└─ Source Type

└─ Owner

|

└─ Title

└─ Content

└─ Section

└─ Metadata

|

└─ Version

└─ Effective Date

└─ Expiry Date

|

└─ Classification

└─ Permission Policy

|

└─ Language
└─ Embedding Reference
└─ Index Reference
|
└─ Freshness
└─ Authority
|
└─ Status

Retrieval Permission Must Happen Before LLM Generation

This deserves explicit acceptance criteria.

Wrong:

Retrieve all tenant data

↓

Send to LLM

↓

Tell LLM:

"Don't mention restricted data."

Correct:

USER

↓

IDENTITY

↓

RBAC / PBAC

↓

AUTHORIZED SCOPE

↓

RETRIEVE ONLY AUTHORIZED DATA

↓

LLM

The LLM should ideally **never receive unauthorized information in the first place.**

RAG vs LLM

Soft Labs should understand the distinction:

LLM

Provides language understanding, reasoning and generation.

RAG

Provides the LLM with relevant **TICVAI-specific/current enterprise knowledge.**

Example:

The LLM may understand generally what a refund is.

It does **not inherently know** Dubai Theme Park's current refund policy.

RAG retrieves:

Dubai Theme Park Refund Policy — Version 7

and gives the relevant approved content to the model.

Then the model explains it.

RAG Does Not Replace APIs

Equally important:

RAG should not become the mechanism for all TICVAI data.

For:

“How many tickets did we sell today?”

use:

Ticketing / Reporting API

not:

Vector Search

This will improve accuracy significantly.

Source Authority

When several sources exist, TICVAI should understand source authority.

Example:

Approved Policy

Higher authority

than:

Internal FAQ

higher than:

Old Training Document

Therefore retrieval ranking can consider:

Relevance + Authority + Freshness + Permission

rather than similarity alone.

Hallucination Handling

When TICVAI does not have enough information, the correct response is:

“I don't have enough approved information to answer that reliably.”

not an invented answer.

For enterprise ticketing, a confident but incorrect answer about:

1. Refunds
2. Finance
3. Access
4. Fraud
5. Pricing

could create serious business problems.

Assistant vs AI Configuration Assistant

This boundary must remain clear.

General AI Assistant

User:

“How does Family Ticket configuration work?”

Assistant:

Explains it.

Configuration Assistant

User:

“Create a Family Ticket for 2 adults + 2 children for AED 450.”

Configuration Assistant:

Interview → Blueprint → Validate → Approval → Execute

So:

General Assistant = Understand / Search / Analyze

while:

Configuration Assistant = Configure / Change

Board 1 vs Board 2

This Board answers:

What does TICVAI AI know and how does it answer?

Board 2 will answer:

What can TICVAI AI actually DO?

That distinction is very important.

Board 2 will therefore cover:

Tool Registry

→ **Capability Discovery**

→ **Agent Planning**

→ **API Invocation**

→ **Action Plan**

→ **Validation**

- **Governance**
- **Approval**
- **Execution**
- **Rollback / Compensation**
- **Audit**

This is where TICVAI moves from an **AI chatbot** to an **AI-powered operational platform**.

Board 1 — Final Screen Map

AI Assistant & Enterprise Knowledge

1. **TICVAI AI Assistant Workspace**
2. **Assistant Profile, Role & Behavior Configuration**
3. **Enterprise Knowledge Source Registry**
4. **Knowledge Ingestion, Processing & Indexing**
5. **Knowledge Permission & Access Control**
6. **Enterprise Search, Retrieval & RAG Configuration**
7. **Grounding, Citations & Answer Reliability**
8. **Conversational Context & Session Intelligence**
9. **Knowledge Quality, Freshness & Feedback Center**
10. **Assistant Testing, Evaluation & Publication**

The complete flow is:

ASK → UNDERSTAND → AUTHORIZE → RETRIEVE → GROUND → REASON → VALIDATE → CITE → ANSWER → EVALUATE

And this establishes the foundation we need before **Board 2 — AI Tools, Agents & Action Orchestration**, which will define how the Assistant safely moves from **answering questions** to **performing controlled actions** inside TICVAI.

TICVAI — Scope of Work

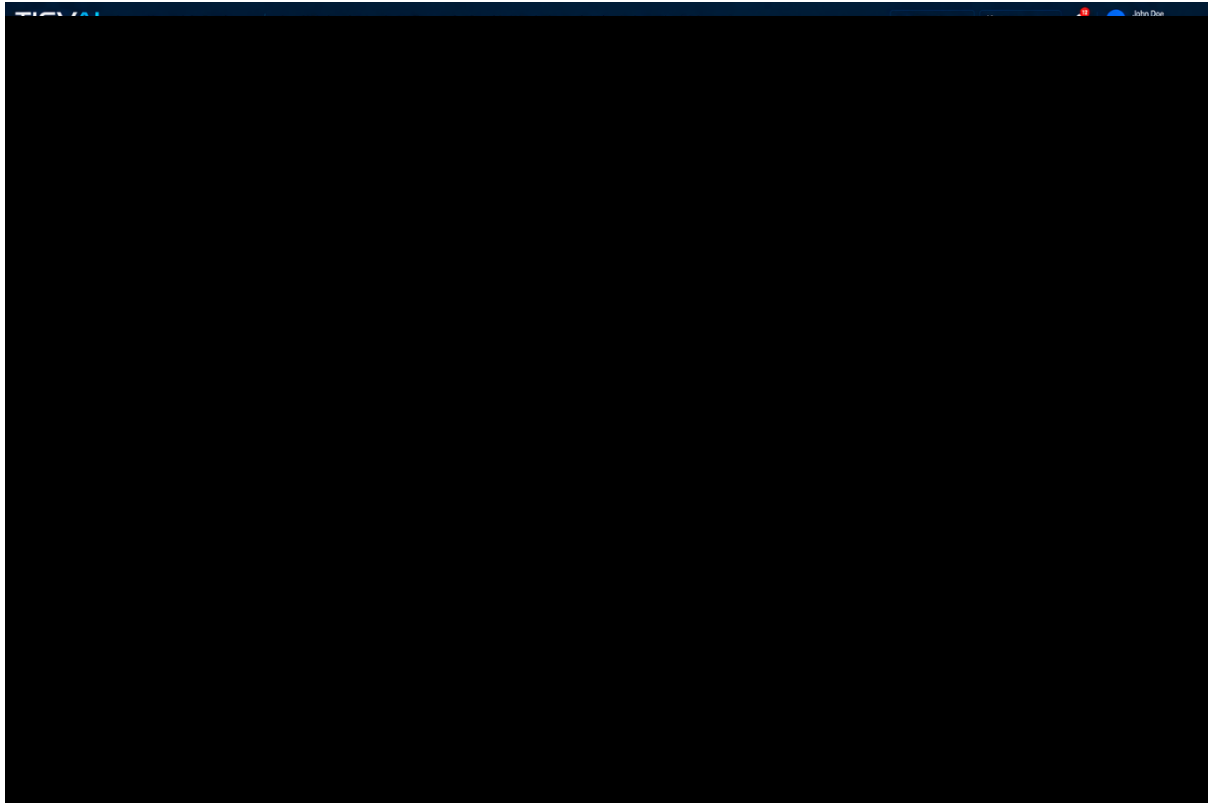
Core AI Platform & Enterprise Intelligence — Board 2: AI Tools, Agents & Action Orchestration

Module: Core AI Platform & Enterprise Intelligence

Board: 2 of 4

Screens: 10

Board Type: AI Agent Orchestration / Tool Registry / Planning / API Invocation / Validation / Governance / Controlled Execution



Board Purpose

Board 1 established:

How TICVAI AI understands questions, retrieves enterprise knowledge, and provides grounded answers.

Board 2 establishes the next level:

How TICVAI AI safely performs controlled actions inside the platform.

This is the layer that converts TICVAI from an AI assistant that only answers questions into an **AI-powered operational platform**.

Example:

User says:

“Create a family ticket for 2 adults and 2 children, valid every Saturday in October, AED 450, available on B2C and POS.”

The AI should not directly modify the database.

Instead:

User Intent

↓

Understand Request

↓

Create Action Plan

↓

Identify Required TICVAI Tools / APIs

↓

Retrieve Current Configuration

↓

Validate Dependencies

↓

Simulate Changes

↓

Apply Governance + RBAC/PBAC

↓

Request Approval where required

↓

Execute through Owning Module APIs

↓

Validate Result

↓

Return Outcome + Audit Trace

The fundamental architecture should be:

AI proposes and orchestrates. TICVAI business services execute.

Core Architecture

USER REQUEST

|



AI ASSISTANT / CONFIGURATION AI

|



INTENT UNDERSTANDING

|



AGENT / PLANNING ENGINE

|

├— Capability Discovery

├— Tool Selection

├— Dependency Resolution

├— Task Planning

└— Parameter Preparation

|



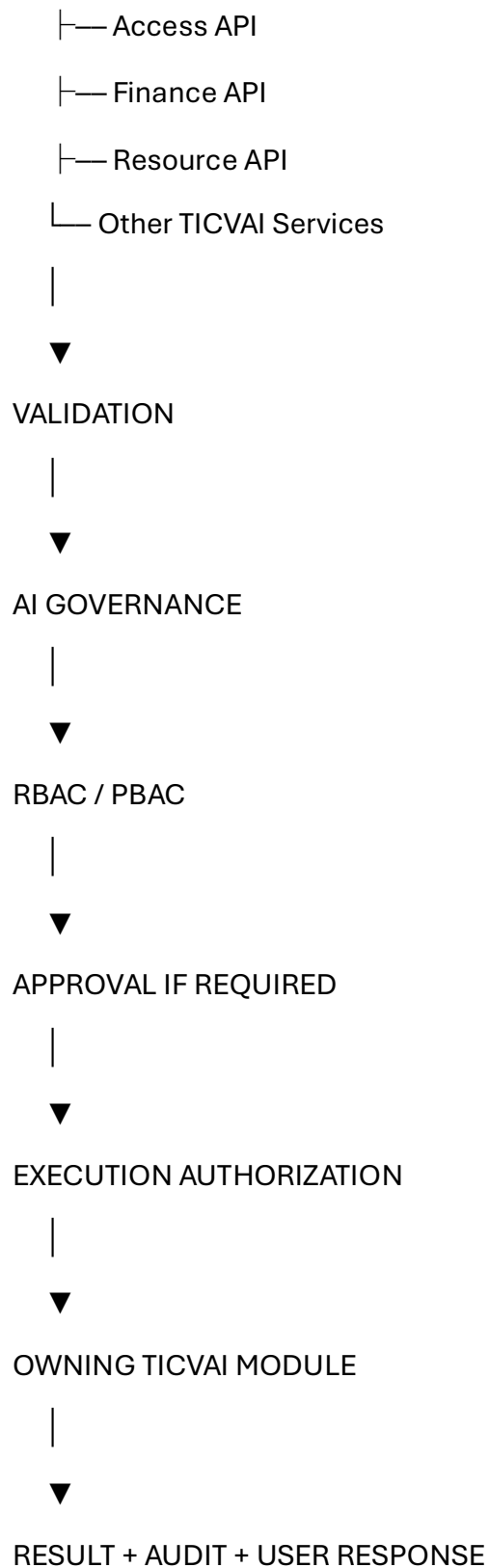
AI TOOL GATEWAY

|

├— Ticketing API

├— Pricing API

├— CRM API



Screen 1 — AI Action & Agent Command Center

Purpose

Provide one central dashboard for monitoring all AI-generated action requests, agent plans, tool executions, approvals and outcomes across TICVAI.

This screen answers:

“What is AI currently trying to do inside TICVAI?”

Header KPIs

1. AI Requests Today
2. Actionable Requests
3. Plans Generated
4. Awaiting Validation
5. Awaiting Approval
6. Currently Executing
7. Completed
8. Failed
9. Blocked by Governance
10. Rolled Back
11. Average Execution Time

Action Status

Statuses:

1. Intent Received
2. Planning
3. Waiting for Information
4. Validation
5. Simulation
6. Awaiting Approval
7. Approved
8. Executing
9. Partially Completed
10. Completed

11. Failed
12. Rolled Back
13. Blocked
14. Cancelled

Action Categories

Examples:

1. Configuration
2. Ticketing
3. Pricing
4. Promotion
5. CRM
6. Finance
7. Access Control
8. Operations
9. Resource
10. F&B / Retail
11. Reporting
12. User Administration

Recent AI Actions Table

Fields:

1. Request ID
2. User
3. Tenant
4. Venue
5. Assistant / Agent
6. Action Type
7. Target Module
8. Risk Level

9. Approval
10. Status
11. Started
12. Duration
13. Outcome

Example

AI-REQ-28931

User Request:

Create weekend family ticket.

Target Modules:

1. Ticketing
2. Pricing
3. Channels

Risk:

Medium

Approval:

Required

Status:

Awaiting Approval

Quick Actions

1. View Active Plans
2. Review Failed Actions
3. Review Awaiting Approval
4. Review Blocked Actions
5. View Execution History
6. Test Agent

Purpose

Maintain the authoritative registry of all TICVAI functions that AI is allowed to discover and potentially invoke.

The AI must never assume it can perform an action simply because the underlying API exists.

Only explicitly registered and governed capabilities should be available.

Tool Categories

Examples:

Ticketing

1. Create Ticket Product
2. Update Ticket Product
3. Publish Ticket
4. Suspend Ticket
5. Retrieve Ticket Configuration

Pricing

1. Retrieve Price
2. Create Price Rule
3. Update Price Rule
4. Run Pricing Simulation

CRM

1. Retrieve Customer
2. Create Segment
3. Create Campaign Draft
4. Add Customer Note

Access Control

1. Retrieve Credential
2. Invalidate Credential
3. Update Access Rule

Finance

1. Retrieve Settlement
2. Retrieve Revenue
3. Generate Reconciliation Report

Tool Record

Each capability should contain:

1. Tool ID
2. Tool Name
3. Owning Module
4. API / Service Endpoint
5. Action Type
6. Read / Write
7. Risk Classification
8. Required Permission
9. Required Approval
10. Environment
11. Tenant Scope
12. Input Schema
13. Output Schema
14. Timeout
15. Retry Policy
16. Idempotency Support
17. Status
18. Version

Example

TOOL-TKT-001

Name:

Create Ticket Product

Owner:

Ticketing

Type:

WRITE

Risk:

Medium

Approval:

Required

API:

Ticket Product Service

Version:

v2.4

Capability Status

1. Active
2. Restricted
3. Test Only
4. Deprecated
5. Disabled

Important

AI should discover tools from this registry rather than from unrestricted API documentation.

Actions

Register Tool | Update Tool | Test | Disable | View Schema | View Governance Policy

Screen 3 — AI Agent & Orchestration Profile Configuration

Purpose

Configure specialized AI agents responsible for different business tasks.

TICVAI should not necessarily use one universal agent for every operation.

Example Agents

Ticket Configuration Agent

Allowed capabilities:

1. Ticketing
2. Pricing
3. Tax
4. Channel
5. Media
6. Access

Operations Agent

Allowed capabilities:

1. Resource
2. Capacity
3. Access
4. Workforce
5. Virtual Queue

Finance Agent

Allowed capabilities:

1. Revenue
2. Settlement
3. Reconciliation
4. Refund Review

Customer Service Agent

Allowed capabilities:

1. Customer
2. Ticket
3. Reschedule
4. Refund Eligibility
5. Membership

Agent Configuration

Fields:

1. Agent ID
2. Name
3. Purpose
4. Description
5. Tenant Scope
6. Venue Scope
7. Allowed Capabilities
8. Restricted Capabilities
9. Maximum Risk Level
10. Planning Mode
11. Approval Profile
12. Governance Profile
13. Model Profile
14. Timeout
15. Max Tool Calls
16. Max Plan Steps
17. Status

Execution Modes

1. Advisory Only
2. Plan Only
3. Prepare Action
4. Execute with Approval
5. Controlled Auto-Execution

These should align with the AI Governance autonomy levels already defined.

Agent Boundary

Example:

Finance Agent should not suddenly modify:

Access Control Gate Rules

unless explicitly authorized.

Actions

Create Agent | Assign Tools | Configure Limits | Test | Activate | Disable

Screen 4 — AI Intent Decomposition & Action Planning Workspace

Purpose

Show how a natural-language request is converted into a structured execution plan before anything is changed.

This is critical for transparency and control.

Example User Request

“Create a family ticket for 2 adults and 2 children for AED 450, valid every Saturday in October, available online and POS.”

AI Interpretation

Structured intent:

Action

Create Ticket Product

Product Type

Family

Entitlement

2 Adults + 2 Children

Price

AED 450

Validity

Every Saturday

Period

1–31 October

Channels

B2C + POS

Missing Information

Example:

1. Refund Policy
2. Capacity
3. Tax Profile
4. Ticket Media
5. Access Rules

The AI may ask follow-up questions if required.

Generated Plan

Example:

1. Retrieve venue calendar
2. Validate October Saturdays
3. Retrieve existing family ticket templates
4. Check pricing/tax rules
5. Create ticket draft
6. Create entitlement definition
7. Assign validity
8. Link channels
9. Validate dependencies
10. Submit for approval
11. Publish after approval

Plan Dependencies

Show:

1. Step Dependency
2. Required Module
3. Required Tool
4. Required Permission

5. Approval Requirement

Plan Status

1. Draft
2. Validated
3. Invalid
4. Needs Input
5. Ready for Simulation
6. Ready for Approval

Actions

Edit Plan | Ask User | Validate Plan | Simulate | Cancel

Screen 5 — AI Tool Invocation & Parameter Builder

Purpose

Translate approved plan steps into controlled, schema-valid API requests.

This screen provides visibility into exactly what TICVAI AI is requesting from each system service.

Example Tool Call

Tool:

Create Ticket Product

Request:

```
{  
  "productType": "FAMILY",  
  "name": "Weekend Family Ticket",  
  "adultQty": 2,  
  "childQty": 2,  
  "currency": "AED",  
  "price": 450,  
  "venueId": "VEN-001"
```

}

Parameter Sources

Each parameter should indicate its source:

1. User Input
2. Current Configuration
3. System Default
4. Policy
5. Retrieved Data
6. Calculated
7. AI Suggested

Example:

Parameter	Value	Source
Currency	AED	Venue Configuration
Adult Qty	2	User Input
Child Qty	2	User Input
Tax Profile	VAT5	Pricing Configuration
Venue	Dubai Theme Park	Session Context

Required Parameter Validation

Highlight:

1. Missing
2. Invalid
3. Conflicting
4. Unauthorized
5. Deprecated

Sensitive Parameters

Masked or protected where required.

Schema Version

AI tool calls must bind to:

1. API Version
2. Schema Version
3. Tool Version

to avoid runtime mismatch.

Actions

Validate Parameters | Preview Request | Test Call | Save Step

Screen 6 — AI Plan Validation, Dependency & Simulation Center

Purpose

Validate the complete AI-generated action before production execution.

This is one of the most important screens in Board 2.

Validation Categories

Business Validation

Example:

1. Ticket valid dates exist
2. Product type supported
3. Price positive
4. Tax profile valid

Configuration Validation

1. Required media assigned
2. Required channel assigned
3. Required access rule exists

Dependency Validation

1. Pricing service available
2. Ticket service available
3. Channel configuration exists

Permission Validation

1. User authorized

2. Agent authorized
3. Tool allowed
4. Scope permitted

Conflict Validation

Example:

Existing ticket has same:

1. Name
2. Validity
3. Product Code

Simulation

Show:

Current State

No product exists.

Proposed State

Weekend Family Ticket

Price:

AED 450

Channels:

B2C + POS

Validity:

4 Saturdays

Impact Preview

1. New Product: 1
2. New Variant: 1
3. Pricing Links: 1
4. Channel Associations: 2
5. Access Policy Association: 1

Validation Result

1. Passed
2. Warning
3. Failed
4. Requires Human Review

Actions

Run Validation | Run Simulation | View Conflicts | Fix Plan | Submit for Approval

Screen 7 — AI Governance, Permission & Approval Gateway

Purpose

Apply the governance decisions already defined in the AI Governance module before execution.

This board does not recreate governance rules.

Instead, it invokes and enforces them.

Effective Permission

Use the existing principle:

Effective AI Permission

=

User Permission

∩

AI Agent Permission

∩

Tool Permission

∩

AI Governance Policy

∩

Owning Module Policy

Example

Action:

Publish New Ticket

User:

Operations Manager

Agent:

Ticket Configuration Agent

Governance:

Approval Required

Owning Module:

Ticketing

Result:

AWAITING APPROVAL

Governance Decision

Possible outputs:

1. Allow
2. Allow with Conditions
3. Prepare Only
4. Approval Required
5. Escalate
6. Block

Approval Package

Include:

1. User Request
2. AI Interpretation
3. Proposed Plan
4. Change Set
5. Validation Results
6. Simulation
7. Risk

8. Impact
9. Required Tools
10. Governance Decision
11. Evidence
12. Version Hash

Approval Binding

Approval must bind to the exact:

Plan Version / Change Set / Hash

If AI modifies the plan materially:

Previous approval becomes invalid.

Actions

View Governance Decision | Submit Approval | Revalidate | Cancel Request

Screen 8 — AI Controlled Execution & Live Orchestration Monitor

Purpose

Execute authorized plans through TICVAI service APIs while monitoring each step in real time.

Execution Header

Example:

Execution EXE-99218

Plan:

Weekend Family Ticket

Steps:

11

Completed:

7

Current:

Create Channel Associations

Status:

Executing

Step Timeline

1. Retrieve Venue Configuration — Completed
2. Validate Dates — Completed
3. Create Ticket Draft — Completed
4. Create Entitlements — Completed
5. Link Pricing — Completed
6. Link Tax — Completed
7. Link B2C — Completed
8. Link POS — Running
9. Link Access Policy — Pending
10. Validate Product — Pending
11. Publish — Pending

Step Details

For each:

1. Tool
2. API
3. Request ID
4. Start Time
5. Duration
6. Result
7. Retry Count
8. Output
9. Error

Execution Controls

Subject to permission:

1. Pause

2. Resume
3. Cancel
4. Retry
5. Escalate

Important

LLM should not be used to decide whether an API technically succeeded.

Success must come from:

Deterministic service response / validation.

Actions

Pause | Retry | View Tool Call | View Response | Escalate

Screen 9 — Failure Handling, Rollback & Compensation Management

Purpose

Define and manage what happens when a multi-step AI action partially fails.

This is essential.

Example:

AI plan:

1. Create Ticket — Success
2. Create Price — Success
3. Link Channels — Success
4. Configure Access — Failed
5. Publish — Not Started

The platform must not simply leave an uncontrolled partial configuration.

Failure Types

1. Validation Failure
2. Permission Failure
3. API Failure
4. Timeout

5. Dependency Failure
6. Conflict
7. Approval Expired
8. Tool Version Mismatch
9. Service Unavailable

Recovery Options

1. Retry Step
2. Skip Step where safe
3. Pause for Human Review
4. Compensating Action
5. Rollback
6. Continue from Checkpoint
7. Cancel Remaining Steps

Compensation

Not every business action can be technically “rolled back.”

Example:

If an external notification was already sent, the system may require:

Compensating Action

rather than rollback.

Rollback Plan

The original plan should define:

1. Reversible Steps
2. Non-Reversible Steps
3. Compensation Method
4. Rollback Order
5. Required Approval

Example

Create Product

Rollback:

Archive Draft Product

Issue Customer Refund

May be:

Non-Reversible

Therefore:

Must require stronger pre-execution approval.

Actions

Retry | Rollback | Execute Compensation | Escalate | Close Failure

Screen 10 — AI Execution History, Audit & Performance Center

Purpose

Provide complete traceability and operational performance analytics for AI actions.

Execution History

Fields:

1. Request ID
2. Trace ID
3. User
4. Agent
5. Intent
6. Plan Version
7. Tool Calls
8. Governance Decision
9. Approval
10. Execution ID
11. Outcome
12. Duration
13. Rollback

14. Final Status

End-to-End Trace

Example:

User Request

↓

Intent INT-882

↓

Plan PLN-441 v3

↓

Validation VAL-119

↓

Governance GOV-893

↓

Approval APR-204

↓

Execution EXE-99218

↓

Tool Calls

↓

Business Objects Created

↓

Final Result

Performance KPIs

1. Total AI Actions
2. Successful Actions
3. Failed Actions
4. Partial Failures
5. Rollbacks

6. Human Intervention
7. Governance Blocks
8. Average Plan Steps
9. Average Tool Calls
10. Average Execution Time
11. Success Rate

Tool Performance

By tool:

1. Call Volume
2. Success Rate
3. Failure Rate
4. Average Latency
5. Timeout Rate
6. Retry Rate

Agent Performance

By agent:

1. Requests
2. Plans Created
3. Validation Failure
4. Approval Rate
5. Execution Success
6. Human Intervention
7. Average Cost

Important

High execution count does not automatically mean a good agent.

Quality must also consider:

1. Correctness
2. Governance Compliance

3. User Outcome
4. Error Rate
5. Rollback Rate

Actions

View Trace | Export Audit | View Performance | Investigate Failure | Compare Agents

Critical Architecture Principles

1. AI Must Never Have Unrestricted Database Write Access

Wrong:

LLM

↓

Database

↓

UPDATE ticket SET ...

Correct:

LLM / Agent

↓

AI Tool Gateway

↓

Governance

↓

RBAC/PBAC

↓

Owning Module API

↓

Business Validation

↓

Database

The business module remains the authoritative owner.

2. Tools Must Be Deterministic

The LLM may decide:

“I need to create a ticket.”

But the actual tool call should be:

CreateTicketProduct()

with:

1. Defined schema
2. Defined permissions
3. Defined API
4. Defined validations
5. Defined response

This provides predictability.

3. AI Planning and Execution Must Be Separate

Do not allow:

AI Thinks → Executes Immediately

Recommended:

Think

↓

Plan

↓

Validate

↓

Simulate

↓

Govern

↓

Approve

↓

Execute

This is especially important for high-impact configuration.

4. Read Tools and Write Tools Should Be Different

Example:

Read

GetTicketProduct

Low risk.

Write

UpdateTicketProduct

Higher risk.

Destructive

DisableTicketProduct

Higher still.

Different actions should therefore have different governance and approval requirements.

5. Idempotency Must Be Supported

Suppose the AI sends:

Create Family Ticket

and the API timeout occurs after creation.

If the AI retries blindly:

TICVAI could create:

Family Ticket

and

Family Ticket Copy

Therefore write operations should support:

Idempotency Key

Example:

AI-REQ-28931-STEP-05

Repeated identical requests should not create duplicate business objects.

6. Execution Should Be Version-Aware

The plan may have been generated using:

Pricing Configuration v7

but before execution someone changes it to:

v8

TICVAI should detect:

Configuration Changed Since Validation

and revalidate the plan.

7. Long-Running Actions Need State Management

Some workflows may last:

1. Seconds
2. Minutes
3. Hours
4. Days

Example:

AI creates a configuration plan today.

Approval comes tomorrow.

The action cannot rely on one LLM conversation remaining open.

The platform must persist:

1. Plan

2. State
 3. Context
 4. Version
 5. Approval
 6. Execution checkpoint
-

8. Human Approval Must Apply to Exact Changes

User approving:

“Create family ticket AED 450”

must not authorize the AI later to create:

“Family Ticket AED 550”

Approval should be bound to:

Exact Change Set Hash

9. Tool Discovery vs Tool Execution

These should be separate concepts.

AI may discover:

“Create Refund”

but governance may decide:

This user cannot execute refunds.

The tool can exist while execution remains blocked.

10. Agent ≠ LLM

Soft Labs should understand:

Agent

is an orchestration construct.

It may use:

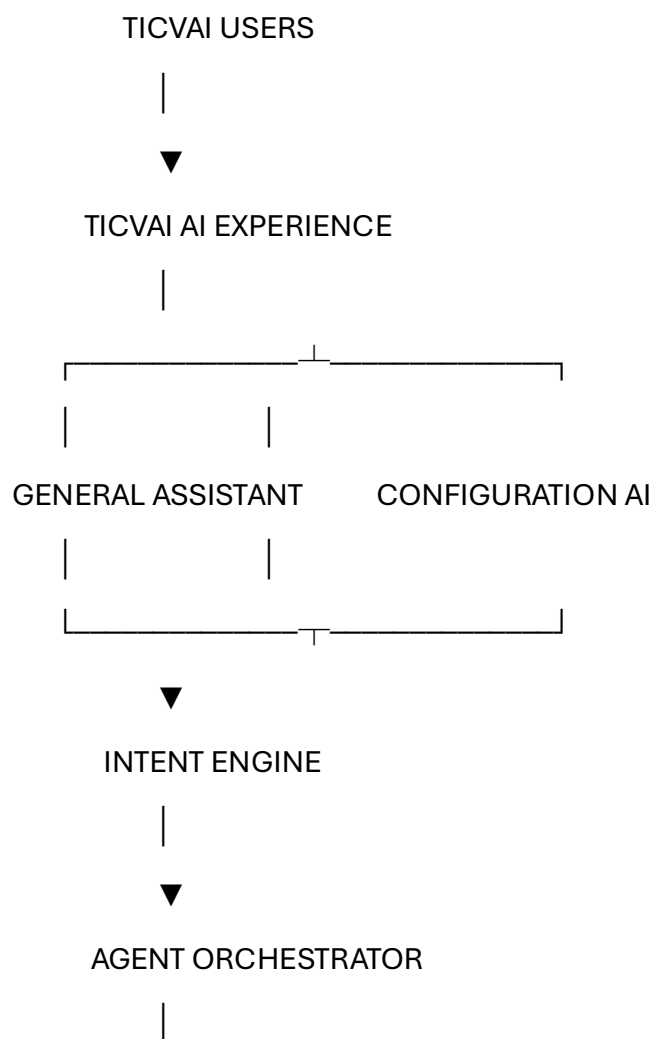
1. LLM

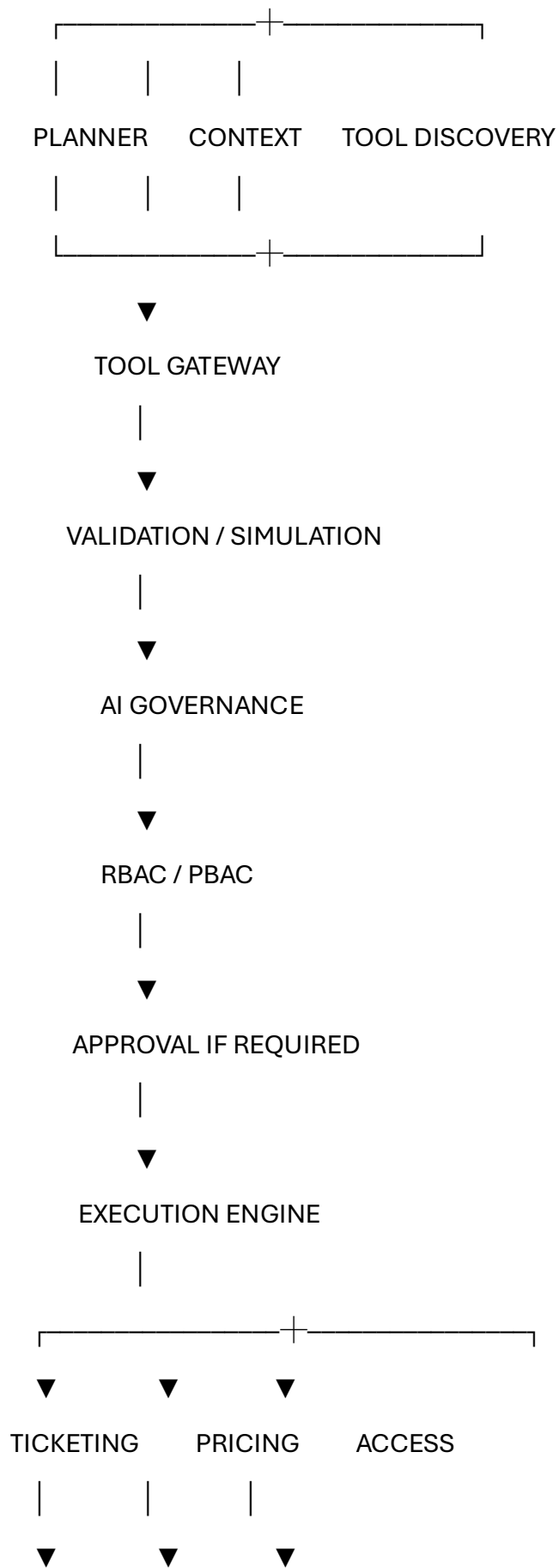
2. Rules
3. APIs
4. Workflow Engine
5. Search
6. Validation
7. Governance
8. Memory
9. Other AI Models

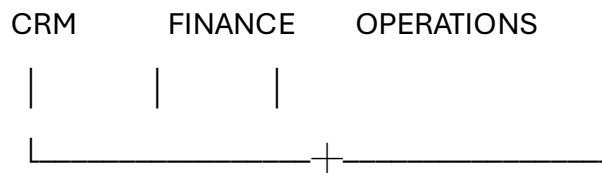
An agent is not simply:

“Another ChatGPT prompt.”

Proposed Technical Architecture







RESULT VALIDATION



AUDIT / TRACE



USER RESULT

Board 1 + Board 2 Relationship

Board 1

ASK



SEARCH



UNDERSTAND



ANSWER

Board 2

REQUEST ACTION



PLAN



TOOLS

↓

VALIDATE

↓

GOVERN

↓

APPROVE

↓

EXECUTE

↓

VERIFY

↓

AUDIT

This gives us a clean separation between:

AI Knowledge

and

AI Action

Example End-to-End Scenario

User:

“Tomorrow is expected to be very busy. Increase the available capacity for the 5 PM Water Park slot by 200 if possible.”

Step 1 — Board 1

AI understands:

1. Tomorrow
2. Water Park
3. 5 PM
4. Capacity

Retrieves:

1. Forecast
2. Existing Capacity
3. Operational Rules

Step 2 — Board 2

AI identifies:

Requested Action: Capacity Change

Step 3

Retrieve:

1. Current capacity = 1,500
2. Physical safe capacity = 1,650

Step 4

AI determines:

Requested:

1,700

But maximum allowed:

1,650

Step 5

Instead of blindly executing:

“I cannot increase the slot by 200 because the maximum permitted capacity is 1,650. I can propose an increase of 150.”

Step 6

User accepts.

Step 7

Plan generated:

1,500 → 1,650

Step 8

Validation passes.

Step 9

Governance requires Operations Manager approval.

Step 10

Approval granted.

Step 11

AI calls:

Capacity Management API

Step 12

Owning module updates capacity.

Step 13

AI verifies:

New Capacity = 1,650

Step 14

Assistant returns:

Capacity successfully increased from 1,500 to 1,650 for the 5 PM slot. The requested increase to 1,700 was not permitted because the configured safe-capacity limit is 1,650.

This is the behavior we want from TICVAI.

Board 2 — Final Screen Map

AI Tools, Agents & Action Orchestration

1. **AI Action & Agent Command Center**
2. **AI Tool & Capability Registry**
3. **AI Agent & Orchestration Profile Configuration**
4. **AI Intent Decomposition & Action Planning Workspace**
5. **AI Tool Invocation & Parameter Builder**
6. **AI Plan Validation, Dependency & Simulation Center**
7. **AI Governance, Permission & Approval Gateway**
8. **AI Controlled Execution & Live Orchestration Monitor**
9. **Failure Handling, Rollback & Compensation Management**

10. AI Execution History, Audit & Performance Center

Complete Board Flow

**INTENT → PLAN → DISCOVER TOOLS → PREPARE PARAMETERS → VALIDATE →
SIMULATE → GOVERN → APPROVE → EXECUTE → VERIFY → RECOVER IF REQUIRED →
AUDIT**

This Board is one of the most important architectural boards in the entire TICVAI AI platform because it defines exactly how AI is allowed to move from **“I understand what you want”** to **“I safely changed the system.”**

TICVAI — Scope of Work

Core AI Platform & Enterprise Intelligence — Board 3: AI Models, LLMs & Provider Management

Module: Core AI Platform & Enterprise Intelligence

Board: 3 of 4

Screens: 10

Board Type: AI Gateway / LLM & Model Registry / Provider Management / Routing / Prompt & Embedding Services / Evaluation / Resilience

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Board Purpose

Board 1 established **what TICVAI AI knows and how it answers**.

Board 2 established **how TICVAI AI plans and safely performs actions**.

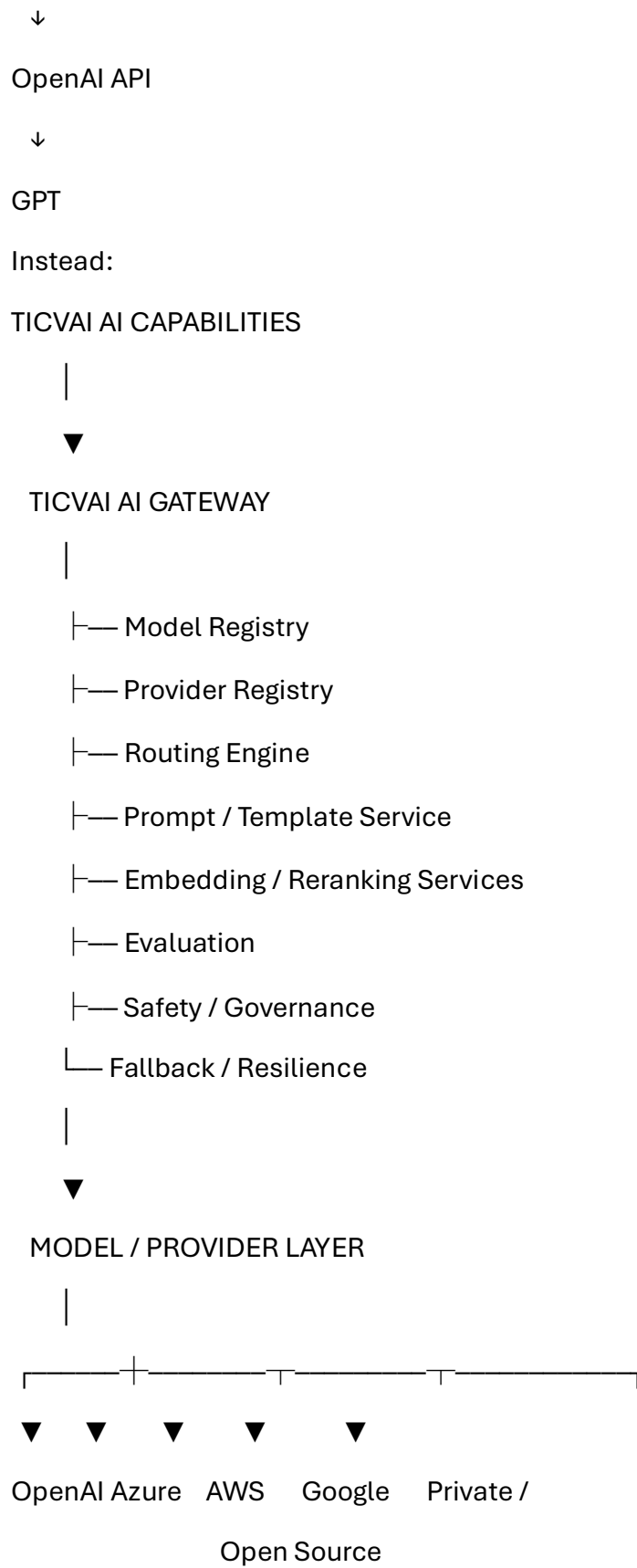
Board 3 establishes the technical intelligence layer underneath them:

Which AI model or provider should TICVAI use for each task, and how does TICVAI manage those models without becoming dependent on one provider?

This is particularly important for Soft Labs.

I do **not** recommend architecture such as:

TICVAI



TICVAI should be able to change a model/provider without redesigning the business modules.

Screen 1 — AI Model & Provider Command Center

Purpose

Provide a central operational view of all AI models, LLMs, embedding models, rerankers and providers configured across TICVAI.

This answers:

“What AI technology is currently powering TICVAI?”

Header KPIs

1. Active Providers
2. Registered Models
3. Production Models
4. LLM Models
5. Embedding Models
6. Reranking Models
7. Specialized ML Models
8. Private Models
9. Healthy Deployments
10. Degraded Deployments
11. Failed Deployments
12. Requests Today

Provider Overview

Examples:

1. OpenAI
2. Azure OpenAI
3. AWS
4. Google
5. Customer-Hosted

6. TICVAI-Hosted
7. Open-Source / Private

Do not hard-code these as mandatory providers. The registry must remain extensible.

Model Categories

1. LLM / Generative
2. Reasoning
3. Embedding
4. Reranking
5. Classification
6. Recommendation
7. Forecasting
8. Fraud / Anomaly
9. Vision
10. Speech
11. Specialized ML

Provider Health

Statuses:

1. Healthy
2. Degraded
3. Rate Limited
4. Unavailable
5. Maintenance
6. Disabled

Active Model Table

Fields:

1. Model
2. Provider
3. Type

4. Deployment
5. Region
6. Environment
7. Assigned Capabilities
8. Version
9. Health
10. Latency
11. Status

Actions

Add Provider | Register Model | View Routing | View Health | Test Model

Screen 2 — AI Provider & Connection Management

Purpose

Configure the AI providers available to TICVAI without exposing provider-specific implementation directly to business modules.

Provider Record

Fields:

1. Provider ID
2. Provider Name
3. Provider Type
4. Deployment Type
5. Endpoint
6. Region
7. Data Residency
8. Authentication Method
9. Secret Reference
10. Network Mode
11. API Version

12. Supported Capabilities
13. Environment
14. Tenant Availability
15. Health Check
16. Status

Deployment Types

Support:

1. Public Cloud API
2. Private Cloud
3. Customer Cloud
4. On-Premise
5. Dedicated Endpoint
6. TICVAI-Hosted
7. Hybrid

Authentication

Credentials must use secure secret management.

Do not store API keys in:

1. Prompt configuration
2. Front-end UI
3. Source code
4. Plain database fields
5. Logs

Instead reference:

Secure Secret / Key Vault

Provider Testing

Allow:

1. Connection Test
2. Authentication Test

3. Model Discovery
4. Latency Test
5. Capability Test
6. Region Test

Data Handling

Display provider characteristics such as:

1. Region
2. Data Processing Location
3. Retention Configuration
4. Private Endpoint Support
5. Customer-Managed Key where applicable
6. Training/Data Usage Configuration where available

Actions

Add Provider | Test Connection | Rotate Secret | Disable | View Models

Screen 3 — AI Model Registry & Model Lifecycle

Purpose

Maintain one authoritative catalog of every model that TICVAI is permitted to use.

A provider and a model are different objects.

Example:

Provider

Azure OpenAI

Model

A particular deployed LLM/version.

Model Record

Fields:

1. Model ID
2. Display Name

3. Provider
4. Provider Model ID
5. Model Type
6. Model Family
7. Version
8. Deployment
9. Region
10. Context Limit
11. Input Types
12. Output Types
13. Streaming Support
14. Tool Calling
15. Structured Output
16. Embedding Dimension where relevant
17. Cost Profile
18. Latency Class
19. Quality Tier
20. Environment
21. Status

Model Lifecycle

Statuses:

Discovered

→ **Registered**

→ **Testing**

→ **Approved**

→ **Production**

→ **Deprecated**

→ **Retired**

Version Management

Example:

Assistant Model

v1 → Production

v2 → Testing

v3 → Evaluation

TICVAI must know exactly which model/version produced an AI result.

Capability Mapping

Example:

Capability	Assigned Model
General Assistant	LLM-A
Configuration AI	Reasoning Model-B
Knowledge Embeddings	Embedding-C
Reranking	Reranker-D
Fraud	Fraud Model-E
Forecasting	Forecast Model-F

Important

The same model does not need to power every AI function.

Actions

Register Model | Clone Configuration | Test | Approve | Promote | Deprecate

Screen 4 — AI Model Routing & Task Assignment Engine

Purpose

Determine which model/provider should handle each AI request.

This is one of the most important technical screens in Board 3.

Instead of application code saying:

Call Provider X Model Y

it should request:

Task:

EnterpriseKnowledgeAnswer

The AI Gateway decides the appropriate model.

Routing Inputs

Routing may consider:

1. AI Capability
2. Task Type
3. Tenant
4. Venue
5. Environment
6. Data Classification
7. Data Residency
8. Required Quality
9. Required Latency
10. Required Context
11. Tool Calling Requirement
12. Structured Output Requirement
13. Language
14. Cost Limit
15. Provider Availability
16. Governance Policy

Example Routing Rule

Task

AI Configuration Planning

Preferred

Reasoning Model A

Fallback

Reasoning Model B

Conditions

1. Tool Calling Required
2. Structured Output Required
3. Region = UAE approved
4. Maximum Latency = configured threshold

Routing Strategies

Support:

1. Fixed Model
2. Priority Routing
3. Weighted Routing
4. Capability-Based
5. Cost-Optimized
6. Latency-Optimized
7. Quality-Optimized
8. Tenant-Specific
9. Data-Residency-Based

Important

Cost optimization must not override:

1. Security
2. Data Residency
3. Governance
4. Capability Requirements

Routing Preview

Input:

Summarize today's executive performance.

Output:

Capability: Executive Assistant

Task: Summarization

Data Classification: Internal

Region: UAE

Selected Route: MODEL-A

Fallback: MODEL-B

Actions

Create Route | Test Route | Simulate | Compare Models | Publish Routing

Screen 5 — Model Fallback, Resilience & Circuit Breaker Management

Purpose

Ensure TICVAI does not become unavailable simply because one AI model or provider experiences a failure.

Failure Conditions

Detect:

1. Provider Unavailable
2. Timeout
3. Rate Limit
4. Authentication Failure
5. Excessive Latency
6. Model Unavailable
7. Invalid Response
8. Structured Output Failure
9. Tool Calling Failure
10. Context Limit Exceeded
11. Safety / Governance Rejection

Fallback Chain

Example:

Primary Model



Failure



Fallback Model 1



Failure



Fallback Model 2



Controlled Degraded Mode

Circuit Breaker

If provider failures exceed configured threshold:

CLOSED

Normal



OPEN

Stop routing new requests



HALF-OPEN

Controlled health testing



CLOSED

Recovered

Fallback Compatibility

The system must verify that fallback models support required features.

Example:

Primary supports:

1. Structured Output
2. Tool Calling
3. Large Context

Fallback does not support Tool Calling.

Therefore it cannot automatically replace the primary model for an action requiring tool invocation.

Degraded Mode

Example:

Configuration Assistant unavailable.

General knowledge search may still operate in:

Search-Only Mode

instead of producing uncontrolled AI responses.

Actions

[Configure Fallback](#) | [Test Failure](#) | [Open Circuit](#) | [Restore Provider](#) | [View Incident](#)

Screen 6 — Prompt, Instruction & AI Template Registry

Purpose

Centrally manage reusable system instructions, prompt templates and structured-output definitions used by TICVAI AI capabilities.

Prompts should not be scattered throughout application source code.

Template Types

1. System Instruction
2. Assistant Behavior
3. Task Prompt
4. Classification Prompt
5. Summarization Prompt
6. Extraction Prompt
7. Planning Prompt
8. Explanation Prompt

9. Structured Output Schema
10. Evaluation Prompt

Prompt Record

Fields:

1. Template ID
2. Name
3. Capability
4. Purpose
5. Owner
6. Version
7. Model Compatibility
8. Language
9. Variables
10. Output Schema
11. Governance Classification
12. Environment
13. Effective Date
14. Status

Example

CONFIG-PLAN-v8

Purpose:

Convert approved business intent into structured configuration plan.

Variables:

1. User Intent
2. Tenant
3. Venue
4. Current Configuration
5. Policy Context

6. Tool Capabilities

Output:

ConfigurationPlan Schema

Versioning

v6 → Superseded

v7 → Production

v8 → Testing

Never silently modify a production prompt.

Testing

Compare:

Prompt v7 + Model A

vs.

Prompt v8 + Model A

vs.

Prompt v8 + Model B

Important

Prompts are application configuration and intellectual property and should have appropriate access controls.

Actions

Create Template | Test | Compare Versions | Approve | Publish | Rollback

Screen 7 — Embedding, Vector Search & Reranking Model Management

Purpose

Manage the models and technical services supporting RAG and semantic retrieval from Board 1.

This should be separate from the knowledge-source configuration itself.

Board 1 determines:

What knowledge exists and how it is retrieved.

Board 3 determines:

Which embedding/reranking technology powers that retrieval.

Embedding Profiles

Fields:

1. Profile ID
2. Embedding Model
3. Provider
4. Dimensions
5. Language Support
6. Input Limit
7. Batch Size
8. Vector Index
9. Distance Metric
10. Tenant Strategy
11. Version
12. Status

Reranking Profiles

Fields:

1. Reranker Model
2. Provider
3. Top-K Input
4. Top-N Output
5. Score Threshold
6. Language
7. Latency Target

Example Flow

User Question

↓

Query Embedding

↓

Vector Search

+

Keyword Search

↓

Candidate Results

↓

Reranker

↓

Best Evidence

↓

LLM

Re-Embedding Impact

Changing an embedding model may require:

Re-embedding existing knowledge.

Show:

1. Documents Affected
2. Chunks Affected
3. Estimated Processing
4. Estimated Cost
5. Index Version

Migration

Support:

Old Index

and

New Index

in parallel during controlled migration.

Actions

Screen 8 — Model Evaluation, Benchmarking & Selection Center

Purpose

Allow TICVAI to scientifically compare models before deciding which should power a capability.

Do not select models simply because:

“This is the newest model.”

The best model depends on the task.

Evaluation Dimensions

Support:

1. Accuracy
2. Groundedness
3. Structured Output Compliance
4. Tool Calling Accuracy
5. Reasoning Quality
6. Retrieval Performance
7. Hallucination Rate
8. Permission Compliance
9. Language Quality
10. Latency
11. Reliability
12. Cost

Evaluation Sets

Examples:

General Assistant

500 questions

Configuration AI

300 configuration scenarios

Tool Calling

200 action scenarios

Arabic

150 Arabic questions

Fraud Explanation

100 cases

Model Comparison

Example:

Metric	Model A	Model B	Model C
Groundedness	96%	94%	91%
Tool Accuracy	98%	93%	N/A
Arabic Quality	94%	97%	90%
Avg Latency	2.1s	1.4s	0.8s
Relative Cost	High	Medium	Low

Weighted Evaluation

For Configuration AI:

- 1. Tool Accuracy = High Importance
- 2. Structured Output = High
- 3. Reasoning = High
- 4. Cost = Medium
- 5. Latency = Medium

For FAQ Assistant:

Cost and latency may receive greater weighting.

Regression

A new model must be tested against the existing production baseline.

Actions

Run Evaluation | Compare | View Failures | Approve Candidate | Assign Model

Screen 9 — Tenant, Region & Deployment Model Policy

Purpose

Allow TICVAI to control which AI models/providers may be used for each tenant, geography and deployment model.

This is essential because TICVAI supports:

1. Cloud
2. Private Cloud
3. On-Premise
4. Hybrid

Policy Dimensions

1. Tenant
2. Region
3. Country
4. Environment
5. Data Classification
6. AI Capability
7. Provider
8. Model
9. Deployment Type
10. Data Residency
11. External Processing Allowed
12. Private Endpoint Required

Example

Tenant A

Cloud AI allowed.

Providers:

A + B.

Government Tenant B

External public AI:

Not Allowed

Required:

Private / Customer-Hosted Model

Tenant C

Sensitive Customer Data:

Only approved regional deployment.

Effective Model Policy

Platform Policy

n

Tenant Policy

n

Regional Policy

n

Data Policy

n

AI Governance

=

Allowed Model Set

Then the routing engine selects only from that set.

Environment Separation

Separate:

1. Development
2. QA
3. UAT
4. Production

A test model should not become available in production simply because it exists in the registry.

Actions

Create Policy | Assign Tenant | Test Policy | View Effective Models | Publish

Screen 10 — AI Model Deployment, Change & Release Management

Purpose

Control how new AI models, provider configurations, prompts and routing policies are introduced into production.

AI changes should follow a governed lifecycle.

Release Objects

A release may include:

1. Model Version
2. Provider Configuration
3. Prompt Version
4. Routing Policy
5. Embedding Profile
6. Reranker
7. Fallback Configuration

Release Flow

Draft

↓

Development

↓

Evaluation

↓

QA

↓

UAT

↓

Approval

↓

Canary

↓

Production

↓

Monitor

Canary Deployment

Example:

New model receives:

5% of eligible AI requests

Old model:

95%

Compare:

1. Accuracy
2. Failures
3. Latency
4. Cost
5. User Feedback

Then:

5%

→ 20%

→ 50%

→ 100%

if approved.

Rollback

If quality deteriorates:

Model v3

↓

Rollback

↓

Model v2

without requiring business modules to change.

This demonstrates the value of the AI Gateway abstraction.

Release Comparison

Show:

Before

Model A

Prompt v7

Route v4

After

Model B

Prompt v8

Route v5

Change Impact

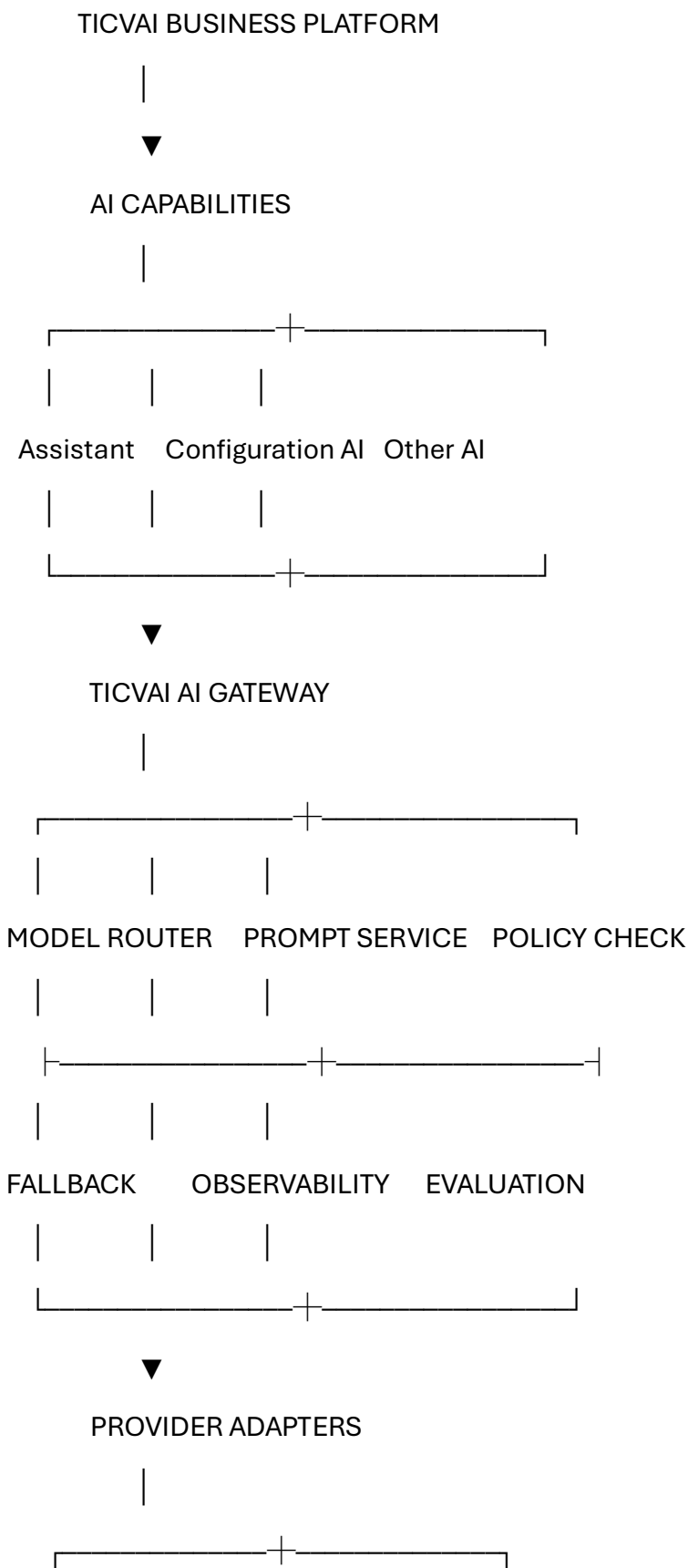
Identify:

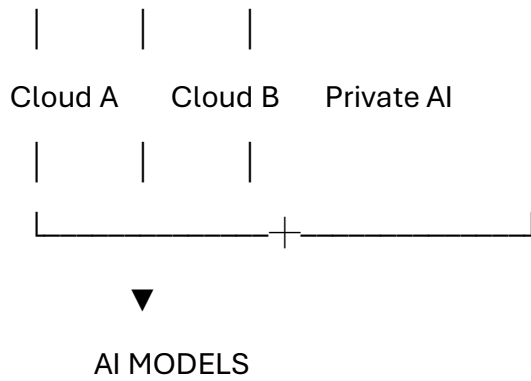
1. Assistant
2. Configuration AI
3. Recommendations
4. Forecasting
5. Fraud
6. Other dependent capabilities

Actions

Create Release | Run Evaluation | Start Canary | Promote | Pause | Rollback

I recommend Soft Labs implement a logical **TICVAI AI Gateway** between TICVAI capabilities and external/internal AI models.





Provider Adapter Pattern

This is another important requirement for Soft Labs.

TICVAI business code should ideally call something similar conceptually to:

AI Gateway

Task:

CONFIGURATION_PLAN

Input:

Business Intent + Context

Requirements:

Structured Output

Tool Calling

High Reasoning

Approved Region

The gateway determines:

which provider and model to call.

Instead of Ticketing or CRM code directly containing provider-specific calls.

This gives TICVAI much greater flexibility later.

Why We Should Not Build Our Own LLM Initially

This board does **not** mean Soft Labs should train a foundational LLM from scratch.

That would require enormous:

1. Training data
2. GPU infrastructure
3. AI research expertise
4. Evaluation
5. Security
6. Maintenance
7. Continuous training investment

It is unnecessary for TICVAI at this stage.

We should build our own:

TICVAI AI Platform

but use suitable external/private models underneath it.

The valuable proprietary layer becomes:

TICVAI Knowledge

- 1.

TICVAI Business Rules

- 1.

Configuration Knowledge Graph

- 1.

TICVAI Tools/APIs

- 1.

AI Orchestration

- 1.

Governance

- 1.

Prompts

1.

Evaluation Data

1.

Business Workflows

That is much more strategically valuable than trying to build another foundational LLM.

LLM Is Only One AI Component

This distinction should be explicitly documented.

TICVAI AI PLATFORM

|

|— LLMs

|

|— Reasoning Models

|

|— Embedding Models

|

|— Rerankers

|

|— Forecasting Models

|

|— Recommendation Models

|

|— Fraud / Anomaly Models

|

|— Rules Engines

|

|— Deterministic Algorithms

Therefore:

AI ≠ LLM

and:

LLM ≠ ChatGPT application.

ChatGPT is an application/product experience; TICVAI would typically integrate suitable underlying model APIs/services through its own AI architecture.

Model Selection Should Be Task-Based

We should not have:

“TICVAI uses Model X.”

We should instead be able to say:

“TICVAI dynamically uses the approved model best suited to each AI capability.”

For example:

Simple Classification

↓

Small / Fast Model

Knowledge Question

↓

General LLM + RAG

Complex Configuration

↓

Reasoning Model

Semantic Retrieval

↓

Embedding + Reranker

Demand Forecast



Forecasting / ML Model

Fraud Detection



Rules + Anomaly + ML

Recommendation



Ranking / Recommendation Model

This is both more technically correct and potentially much more cost-efficient.

Model Response Must Not Be Treated as Truth

Another critical acceptance principle:

MODEL OUTPUT

≠

SYSTEM FACT

For structured operations:

Model Output



Schema Validation



Business Validation



Governance



TICVAI API

For knowledge answers:

Model Output

↓

Grounding / Evidence

↓

Citation / Validation

↓

Response

Board 3 Integration With Existing AI Modules

AI Configuration Assistant

Uses:

1. Reasoning models
2. Structured output
3. Tool calling
4. Configuration knowledge

Recommendation AI

May use:

1. Ranking
2. Embeddings
3. ML
4. Contextual models
5. Rules

It should not depend exclusively on an LLM.

Forecasting AI

Uses:

1. Time-series
2. Statistical models

3. ML

LLM may explain results, but should not be the forecasting engine itself.

Fraud AI

Uses:

1. Rules
2. Velocity
3. Statistical detection
4. ML
5. Graph analysis
6. Anomaly detection

LLM can assist with explanation/investigation summaries.

General Assistant

Uses:

1. LLM
2. RAG
3. Structured data
4. Tools

This keeps the overall TICVAI AI architecture technically sound.

Board 3 — Final Screen Map

AI Models, LLMs & Provider Management

1. **AI Model & Provider Command Center**
2. **AI Provider & Connection Management**
3. **AI Model Registry & Model Lifecycle**
4. **AI Model Routing & Task Assignment Engine**
5. **Model Fallback, Resilience & Circuit Breaker Management**
6. **Prompt, Instruction & AI Template Registry**
7. **Embedding, Vector Search & Reranking Model Management**

8. **Model Evaluation, Benchmarking & Selection Center**
9. **Tenant, Region & Deployment Model Policy**
10. **AI Model Deployment, Change & Release Management**

Board 3 Complete Flow

REGISTER PROVIDER → REGISTER MODEL → CLASSIFY CAPABILITY → EVALUATE → APPROVE → DEFINE ROUTING → CONFIGURE FALLBACK → ASSIGN TENANT/REGION POLICY → RELEASE → MONITOR → CHANGE / ROLLBACK

After this, **Board 4 — AI Platform Operations, Consumption & ModelOps** should complete the Core AI Platform by covering the operational side: **AI consumption and token/compute usage, cost allocation, budgets and quotas, latency, availability, observability, model/runtime performance, AI service health, capacity, incidents, SLAs, optimization and production ModelOps.**

TICVAI — Scope of Work

Core AI Platform & Enterprise Intelligence — Board 4: AI Platform Operations, Consumption & ModelOps

Module: Core AI Platform & Enterprise Intelligence

Board: 4 of 4

Screens: 10

Board Type: AI Operations / Consumption / FinOps / Observability / SLA / Capacity / ModelOps / Incident Management / Optimization

 The picture can't be displayed.

Board Purpose

Boards 1–3 established:

Board 1 — AI Assistant & Enterprise Knowledge

How TICVAI AI knows, retrieves, grounds and answers.

Board 2 — AI Tools, Agents & Action Orchestration

How AI plans and safely performs actions.

Board 3 — AI Models, LLMs & Provider Management

Which models/providers TICVAI uses and how they are routed, evaluated and released.

Board 4 answers the final operational question:

“Once all of this AI is running in production, how do we operate, monitor, control, scale and pay for it?”

This board must give TICVAI visibility from:

Platform → Tenant → Venue → AI Capability → Agent → Provider → Model → Request

so we can understand exactly:

1. Who is consuming AI?
2. Which capability is consuming it?
3. Which model/provider is being used?
4. How much does it cost?
5. Is performance acceptable?
6. Are budgets being exceeded?
7. Are providers healthy?
8. Is capacity sufficient?
9. Are failures increasing?
10. Which models are producing poor results?
11. Are we meeting SLA/SLO targets?
12. What needs optimization?

The operational architecture is:

TICVAI AI REQUESTS

|



AI GATEWAY

|

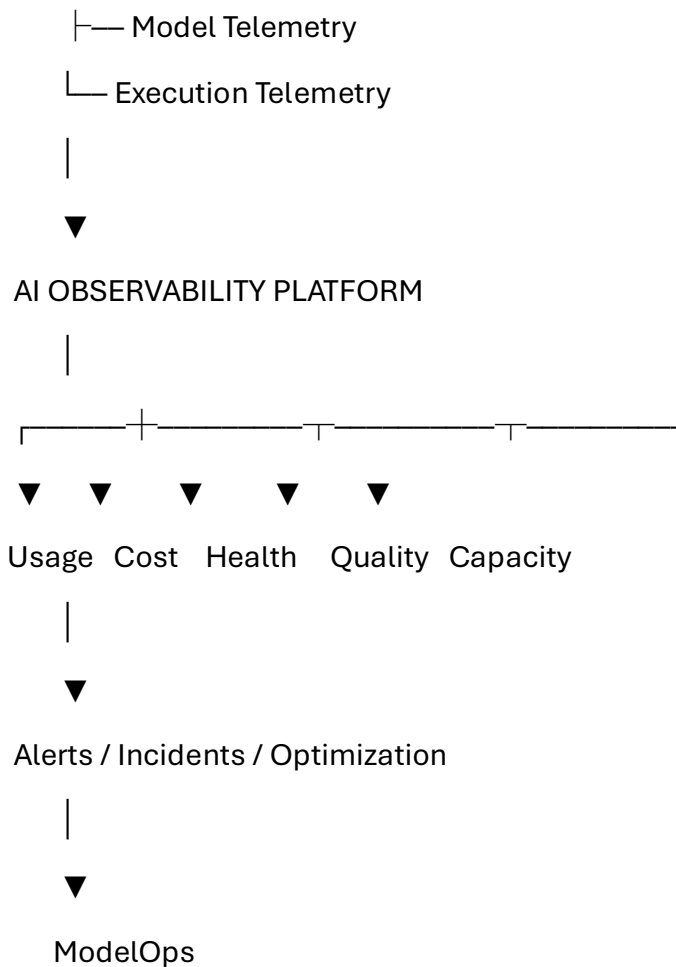
└— Usage Telemetry

└— Cost Telemetry

└— Performance Telemetry

└— Quality Telemetry

└— Provider Telemetry



Screen 1 — AI Operations Command Center

Purpose

Provide the central operational dashboard for the entire TICVAI AI platform.

This should answer:

“Is TICVAI AI healthy right now?”

Header KPIs

1. AI Requests Today
2. Active Tenants
3. Active AI Capabilities
4. Active Models
5. Active Providers
6. Successful Requests

7. Failed Requests
8. Degraded Requests
9. Average Latency
10. AI Cost Today
11. Open Alerts
12. Active Incidents
13. SLA Compliance

AI Platform Health

Overall:

Healthy / Degraded / Critical

Break down by:

1. AI Assistant
2. Configuration AI
3. Recommendations
4. Forecasting
5. Fraud
6. Agents / Actions
7. Knowledge / RAG
8. Embeddings
9. Model Gateway

Request Trend

Show:

1. Total Requests
2. Successful
3. Failed
4. Fallback
5. Throttled

by:

1. Hour
2. Day
3. Week
4. Month

Provider Health

Example:

Provider	Health	Requests	Latency	Error
Provider A	Healthy	48K	1.8s	0.3%
Provider B	Healthy	21K	1.2s	0.2%
Private AI	Degraded	8K	3.4s	4.8%

Operational Alerts

Examples:

1. Latency increased 38%
2. Provider approaching rate limit
3. Configuration AI error rate above SLO
4. Tenant approaching AI budget
5. Embedding queue backlog increasing

Actions

[View Health](#) | [View Alerts](#) | [View Incidents](#) | [View Consumption](#) | [View Providers](#)

Screen 2 — AI Consumption & Usage Analytics

Purpose

Provide detailed measurement of AI consumption throughout TICVAI.

Consumption must not be tracked only by provider.

We need attribution to the actual business use.

Consumption Hierarchy

Platform

↓

Tenant



Venue



AI Capability



Agent / Assistant



Task



Provider



Model

Usage Metrics

For LLMs:

1. Requests
2. Input Tokens
3. Output Tokens
4. Cached Tokens where supported
5. Total Tokens
6. Context Size
7. Tool Calls

For embeddings:

1. Embedding Requests
2. Tokens / Characters Processed
3. Documents
4. Chunks

For other models:

1. Inference Calls
2. Compute Time
3. Batch Jobs
4. GPU/CPU Consumption where applicable

Consumption by Capability

Examples:

1. General Assistant
2. Configuration Assistant
3. Recommendations
4. Fraud
5. Forecasting
6. Knowledge/RAG
7. AI Agents

Consumption by Tenant

Table:

1. Tenant
2. Requests
3. Tokens / Compute
4. Cost
5. Budget Utilization
6. Growth
7. Status

Drilldown

Example:

Tenant A

- Dubai Theme Park
- AI Configuration
- Configuration Planning

→ Model A

→ 12,480 requests

Usage Anomaly Detection

Examples:

Configuration AI token consumption increased 280% today.

or:

Tenant XYZ usage is 4.2× its normal daily baseline.

This should create an investigation signal, not automatically assume abuse.

Actions

Drill Down | Compare Period | Export Usage | View Anomaly

Screen 3 — AI Cost, Budget & FinOps Management

Purpose

Give TICVAI commercial control over AI infrastructure consumption.

This becomes increasingly important as AI usage scales.

Cost KPIs

1. Cost Today
2. Cost MTD
3. Forecast Monthly Cost
4. Budget
5. Budget Remaining
6. Cost per AI Request
7. Cost per Tenant
8. Cost per Capability
9. Cost per Provider
10. Cost per Model

Cost Allocation

Allocate costs by:

1. Tenant
2. Venue
3. Module
4. AI Capability
5. Agent
6. Model
7. Provider
8. Environment

Example

Capability	Requests	Cost	Cost/Request	% Total
Assistant	120K	\$820	\$0.0068	28%
Configuration	38K	\$960	\$0.0253	33%
Recommendations	420K	\$480	\$0.0011	16%
Fraud	210K	\$390	\$0.0019	13%

Budget Configuration

Support:

1. Platform Budget
2. Tenant Budget
3. Capability Budget
4. Environment Budget

Thresholds

Example:

1. 70% — Informational
2. 85% — Warning
3. 95% — Critical
4. 100% — Budget Policy

Budget Action Policy

At threshold, configurable actions may include:

1. Notify
2. Escalate
3. Recommend cheaper route
4. Apply quota where contractually/governance permitted
5. Restrict non-critical AI
6. Require approval for additional consumption

Do **not** silently disable critical AI capabilities solely because a budget threshold was reached unless explicitly configured.

Cost Forecast

Use current consumption trend to estimate:

Projected Month-End AI Cost

Clearly label it as forecast.

Actions

Set Budget | View Cost Drivers | Forecast Cost | Export | Review Optimization

Screen 4 — AI Quota, Rate Limit & Consumption Policy

Purpose

Control how much AI capacity different tenants, capabilities and users are permitted to consume.

This is different from cost reporting.

Quota Dimensions

Support quotas by:

1. Tenant
2. Venue
3. User / Role where appropriate
4. AI Capability
5. Agent
6. Provider
7. Model

8. API
9. Environment

Quota Types

1. Requests / Minute
2. Requests / Hour
3. Requests / Day
4. Tokens / Day
5. Tokens / Month
6. Compute Units
7. Concurrent Requests
8. Agent Executions
9. Tool Calls
10. Embedding Jobs

Example

Tenant A

General Assistant:

100,000 requests/month

Configuration AI:

20,000 requests/month

Concurrent Agent Executions:

50

Rate Limits

Support:

1. Soft Limit
2. Hard Limit
3. Burst Limit
4. Sustained Limit

Priority Classes

Examples:

P1 — Critical

Fraud / security-critical processing

P2 — Operational

Configuration / operational AI

P3 — Standard

Employee Assistant

P4 — Background

Batch knowledge processing

During constrained capacity, TICVAI can prioritize according to policy.

Quota Handling

When limit reached:

1. Warn
2. Queue
3. Throttle
4. Route to alternative model
5. Degrade capability
6. Block non-critical request
7. Escalate

according to configuration.

Actions

Create Quota | Adjust Limit | Test Policy | View Utilization | Override Temporarily

Screen 5 — AI Performance, Latency & Reliability Monitoring

Purpose

Monitor the technical performance of the AI platform end-to-end.

Provider latency alone is insufficient.

A user request may involve:

Gateway + Retrieval + Reranking + LLM + Tools + Validation

Each component must be measurable.

End-to-End Latency

Break down:

Total Response Time

|

|— Gateway

|— Authentication

|— Permission

|— Retrieval

|— Reranking

|— Model

|— Tool Calls

|— Validation

|— Response Processing

Metrics

1. P50 Latency
2. P95 Latency
3. P99 Latency
4. Throughput
5. Success Rate
6. Error Rate
7. Timeout Rate
8. Retry Rate
9. Fallback Rate
10. Queue Time

Performance by Capability

Example:

Capability	P50	P95	Success	Fallback
Assistant	1.7s	4.1s	99.7%	1.2%
Configuration	4.8s	11.2s	99.2%	2.4%
Fraud	0.18s	0.42s	99.99%	0.1%

Dependency Trace

Example:

Request slow because:

Vector Search 0.2s

Reranking 0.4s

LLM 7.8s

Therefore:

Primary bottleneck = Model inference.

Actions

[View Trace](#) | [Compare Models](#) | [View Slow Requests](#) | [Analyze Bottleneck](#)

Screen 6 — AI Service Health, SLA/SLO & Availability Management

Purpose

Define and monitor service-level objectives for different TICVAI AI capabilities.

Not every AI capability requires the same availability.

Service Catalog

Examples:

1. General Assistant
2. Configuration AI
3. Fraud Screening
4. Recommendation Engine
5. Forecasting
6. Knowledge Retrieval
7. Embedding Service

8. AI Gateway

SLO Configuration

Per service:

1. Availability Target
2. Latency Target
3. Error Target
4. Maximum Timeout
5. Recovery Target
6. Data Freshness where applicable

Example

Fraud Screening

Availability:

99.99%

P95:

< 300ms

because fraud screening may be in a transaction path.

Executive AI Assistant

Availability:

99.5%

P95:

< 5 sec

Different operational requirement.

Error Budget

Track:

1. SLO Target
2. Actual
3. Error Budget
4. Error Budget Consumed

5. Remaining

Important

An SLO breach should not automatically mean the underlying provider breached its contractual SLA.

These are separate concepts.

Availability Timeline

Show:

1. Healthy
2. Degraded
3. Partial Outage
4. Unavailable

Actions

[Configure SLO](#) | [View Breach](#) | [View Availability](#) | [Generate SLA Report](#)

Screen 7 — AI Capacity, Scaling & Workload Management

Purpose

Ensure TICVAI AI has sufficient capacity for expected workloads and traffic peaks.

This is especially important for large venues where demand can spike around:

1. Ticket releases
2. Major events
3. Opening time
4. Campaign launches
5. High-volume entry
6. Batch processing

Capacity Metrics

1. Current Requests/sec
2. Peak Requests/sec
3. Concurrent Requests

4. Queue Depth
5. Worker Utilization
6. GPU/CPU Utilization where relevant
7. Provider Quota Utilization
8. Embedding Queue
9. Agent Execution Queue

Capacity by Workload

Examples:

Interactive

Assistant conversations.

Real-Time

Fraud / recommendation.

Transactional

AI action orchestration.

Batch

Knowledge indexing / forecasting / evaluation.

Workload Isolation

Batch embedding jobs should not consume all resources needed for real-time fraud screening.

Support separate:

1. Queues
2. Workers
3. Priorities
4. Capacity pools

where architecture requires.

Scaling Policy

Support:

1. Minimum Capacity

2. Maximum Capacity
3. Auto-Scaling Threshold
4. Queue Threshold
5. Scale-Up Rule
6. Scale-Down Rule
7. Cooldown

Capacity Forecast

Estimate expected capacity requirement from historical usage and known events.

Actions

View Capacity | Configure Scaling | Rebalance Workload | Run Load Test

Screen 8 — AI Alert, Incident & Operational Response Center

Purpose

Centralize AI platform operational alerts and incidents.

This is distinct from **AI Governance incidents**.

Governance incident:

AI violated or potentially violated a policy.

Operational incident:

AI provider is unavailable or performance is degraded.

They can be linked when both apply.

Alert Categories

1. Provider Outage
2. Model Failure
3. Latency
4. Error Rate
5. Rate Limit
6. Quota
7. Cost

8. Capacity
9. Embedding Failure
10. Retrieval Failure
11. Agent Failure
12. Tool Failure
13. Deployment Failure
14. SLO Breach

Alert Severity

1. Critical
2. High
3. Medium
4. Low
5. Informational

Alert Correlation

Example:

200 individual model timeout alerts

may correlate into:

Provider A Degradation Incident

instead of creating 200 separate incidents.

Incident Record

Fields:

1. Incident ID
2. Severity
3. Service
4. Provider
5. Model
6. Tenant Impact
7. Capability Impact

8. Start Time
9. Detection
10. Owner
11. Status
12. Root Cause
13. Mitigation
14. Resolution
15. Related Alerts

Incident Lifecycle

Detected

→ **Acknowledged**

→ **Investigating**

→ **Mitigating**

→ **Monitoring**

→ **Resolved**

→ **Post-Incident Review**

Response Options

1. Activate Fallback
2. Disable Route
3. Open Circuit Breaker
4. Shift Traffic
5. Reduce Batch Load
6. Increase Capacity
7. Notify Stakeholders

subject to authorization.

Actions

Acknowledge | Assign | Mitigate | Activate Fallback | Resolve | Start Review

Screen 9 — AI ModelOps, Runtime Quality & Production Monitoring

Purpose

Monitor whether deployed AI models continue performing appropriately after release.

Board 3 evaluates a model **before and during deployment**.

Board 4 monitors it **continuously in production**.

ModelOps Monitoring

Track:

1. Model Version
2. Deployment
3. Request Volume
4. Error Rate
5. Latency
6. Fallback
7. Output Validation
8. Quality Indicators
9. User Feedback
10. Grounding Failures
11. Structured Output Failures
12. Tool Calling Failures

Runtime Quality by AI Type

For Assistant:

1. Groundedness
2. Citation Availability
3. Unanswered Rate
4. Negative Feedback

For Configuration AI:

1. Plan Validation Pass Rate
2. User Correction Rate

3. Execution Success

4. Rollback Rate

For Recommendations:

1. Recommendation Quality

2. Acceptance / Engagement

3. Experiment Results

For Forecasting:

1. Forecast Error

2. Bias

3. Interval Coverage

For Fraud:

1. Confirmed Outcomes

2. False Positives

3. Detection Performance

Important

Do not create one universal:

“AI Accuracy = 94%”

for all TICVAI AI.

Each AI capability requires appropriate quality metrics.

Production Comparison

Compare:

Model v3

vs.

Model v2

on:

1. Quality

2. Latency

3. Cost

4. Errors
5. User Feedback

Drift / Degradation Signals

Examples:

1. Grounding quality declining
2. Tool-call validation failures increasing
3. Forecast bias increasing
4. Fraud false-positive rate changing
5. Recommendation performance deteriorating

Actions

**Investigate Model | Compare Version | Run Evaluation | Recommend Rollback |
Create Improvement Candidate**

Screen 10 — AI Operations Optimization & Management Reporting

Purpose

Turn operational telemetry into controlled optimization opportunities and management reporting.

This closes the operational AI lifecycle.

Executive AI Operations KPIs

1. Total AI Requests
2. AI Cost
3. Cost per Request
4. AI Availability
5. AI Success Rate
6. P95 Latency
7. SLA/SLO Compliance
8. Fallback Rate
9. Active Incidents
10. Production Models

11. Budget Utilization
12. Optimization Opportunity

Optimization Categories

Cost

Example:

38% of Assistant requests are simple classification/summarization tasks currently routed to a high-cost reasoning model.

Recommendation:

Evaluate smaller model.

Performance

Model B is 42% faster for Arabic FAQ requests with equivalent evaluated quality.

Reliability

Provider B has lower timeout rate for the selected regional workload.

Retrieval

Reranking contributes 620ms average latency but improves grounded-answer quality by 9%.

This creates a tradeoff decision rather than automatically removing it.

Optimization Candidate

Each candidate should include:

1. Current Configuration
2. Proposed Change
3. Reason
4. Expected Benefit
5. Cost Impact
6. Performance Impact
7. Quality Impact
8. Risk
9. Evidence
10. Evaluation Required

11. Approval Required

Optimization Lifecycle

Observe

↓

Analyze

↓

Identify Opportunity

↓

Create Candidate

↓

Evaluate

↓

Simulate / Benchmark

↓

Governance Review

↓

Approval

↓

Deploy through Board 3

↓

Measure

Reports

Support:

1. Daily AI Operations
2. Weekly AI Performance
3. Monthly AI Consumption
4. Tenant Consumption
5. AI Cost

6. Provider Performance
7. Model Performance
8. SLO/SLA
9. Incident
10. Optimization

Actions

Generate Report | Review Opportunities | Create Optimization Plan | Send to Model Release

Critical Distinction — Board 3 vs Board 4

Soft Labs should clearly understand this boundary.

Board 3 — AI Models & Providers

Answers:

What should TICVAI use?

It handles:

Provider → Model → Evaluation → Routing → Fallback Configuration → Release

Board 4 — AI Operations & ModelOps

Answers:

How is it actually performing in production?

It handles:

Usage → Cost → Performance → Reliability → Capacity → Incident → Runtime Quality → Optimization

Therefore:

BOARD 3

Configure & Deploy

↓

PRODUCTION

↓

BOARD 4

Observe & Operate



Improvement Candidate



BOARD 3

Evaluate & Release

This creates a continuous operational loop.

Critical Requirement — One AI Telemetry Standard

I strongly recommend that Soft Labs does not let each AI module invent its own monitoring structure.

Every AI request should create a standardized telemetry envelope.

AI Request Telemetry Object

AIRequestTelemetry

|

├— Request ID

├— Trace ID

├— Timestamp

|

├— Tenant

├— Venue

├— Environment

|

├— User / Service Context

|

├— AI Capability

├— Agent

├— Task Type

|

└─ Provider

└─ Model

└─ Model Version

└─ Route

|

└─ Prompt / Template Version

|

└─ Input Consumption

└─ Output Consumption

└─ Cached Consumption

|

└─ Tool Calls

└─ Retrieval Calls

|

└─ Latency

└─ Queue Time

|

└─ Status

└─ Error Type

└─ Retry Count

└─ Fallback

|

└─ Estimated / Actual Cost

|

└─ Governance Reference

└─ Execution Reference

|

└─ Quality / Outcome References

This allows TICVAI to trace one request from:

User → Assistant → RAG → Model → Tool → Business API → Result

instead of having disconnected logs.

Critical Requirement — Distributed Trace ID

The same **Trace ID** should travel through the entire transaction:

User

↓

AI Assistant

↓

AI Gateway

↓

RAG

↓

Model

↓

Agent

↓

Tool

↓

TICVAI API

↓

Business Service

Then operations can answer:

“Why did this AI request take 14 seconds?”

and see:

Authentication	80ms
Permission	40ms
Knowledge Retrieval	420ms
Reranking	610ms
Model	8.2s
Tool Call	3.1s
Validation	240ms
Response	180ms

That will be extremely valuable during production troubleshooting.

Critical Requirement — AI Cost Attribution

I would also make this an explicit acceptance criterion.

It is not sufficient to know:

OpenAI invoice = \$80,000

TICVAI should be capable of understanding:

\$80,000

- |
- |— Tenant A \$30,000
- |— Tenant B \$24,000
- |— Tenant C \$18,000
- |— Internal \$8,000

Then:

Tenant A

- |
- |— Assistant \$8,000
- |— Configuration \$10,000
- |— Recommendations \$5,000
- |— Forecasting \$3,000

└ Other \$4,000

This is important because later TICVAI may choose to price AI through:

1. Included AI allowance
2. Usage-based pricing
3. AI credits
4. Premium AI package
5. Enterprise unlimited/fair-use
6. Tenant-specific commercial agreements

The Subscription & Licensing module should remain the owner of commercial charging, while this Board provides the authoritative **AI usage and cost telemetry**.

Critical Requirement — AI Must Not Block Core Ticketing Unnecessarily

This is especially important for TICVAI.

If an AI provider fails, the entire venue should not stop selling tickets unless the AI capability is genuinely mandatory for that specific transaction.

Example:

Recommendation AI unavailable

Checkout should continue:

without recommendations.

General Assistant unavailable

POS continues operating.

Forecasting unavailable

Ticket sales continue using existing operational configuration.

Configuration AI unavailable

Admin can use traditional manual configuration.

Fraud AI unavailable

Behavior depends on the defined risk/failover policy because fraud may be transaction-sensitive.

Therefore every AI capability should define a:

Failure / Degradation Mode

Examples:

1. Continue Without AI
2. Use Cached Result
3. Use Rules-Only
4. Use Fallback Model
5. Manual Process
6. Queue Request
7. Fail Closed

based on the capability's risk.

ModelOps Does Not Mean Automatic Model Changes

Board 4 may detect:

Model B is cheaper and faster.

It should not automatically switch production because of that alone.

The correct loop is:

Production Observation

↓

Optimization Candidate

↓

Evaluation

↓

Benchmark

↓

Governance

↓

Approval

↓

Board 3 Release Management

↓

Canary

↓

Production

This maintains control.

Board 4 — Final Screen Map

AI Platform Operations, Consumption & ModelOps

1. **AI Operations Command Center**
2. **AI Consumption & Usage Analytics**
3. **AI Cost, Budget & FinOps Management**
4. **AI Quota, Rate Limit & Consumption Policy**
5. **AI Performance, Latency & Reliability Monitoring**
6. **AI Service Health, SLA/SLO & Availability Management**
7. **AI Capacity, Scaling & Workload Management**
8. **AI Alert, Incident & Operational Response Center**
9. **AI ModelOps, Runtime Quality & Production Monitoring**
10. **AI Operations Optimization & Management Reporting**

Complete Board 4 Flow

OBSERVE → MEASURE USAGE → ATTRIBUTE COST → CONTROL QUOTA → MONITOR PERFORMANCE → MEASURE SLO → MANAGE CAPACITY → DETECT INCIDENT → MONITOR MODEL QUALITY → OPTIMIZE → EVALUATE → RELEASE

With Board 4, the **Core AI Platform & Enterprise Intelligence module becomes complete at 4 boards / 40 screens:**

Board 1 — AI Assistant & Enterprise Knowledge

Board 2 — AI Tools, Agents & Action Orchestration

Board 3 — AI Models, LLMs & Provider Management

Board 4 — AI Platform Operations, Consumption & ModelOps

Together, these four boards give Soft Labs much more than “integrate an LLM.” They define the actual **TICVAI enterprise AI platform architecture**.